Assembly manual
for the frame of PNPv3

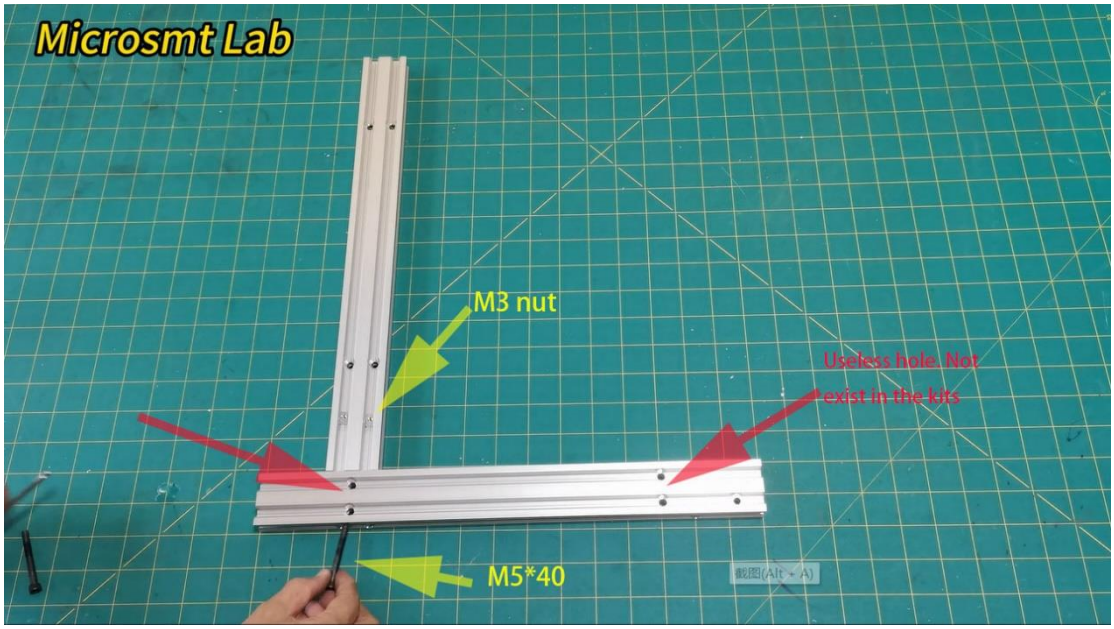
2024.12.26

Please also refer to the assembly demonstration video. Can be completed faster.

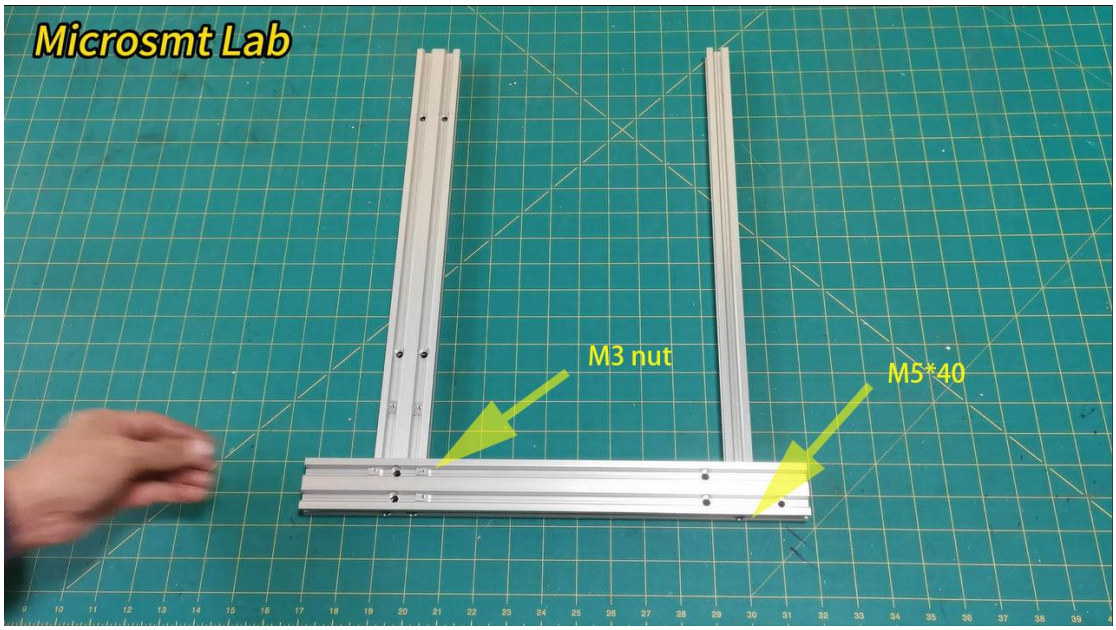
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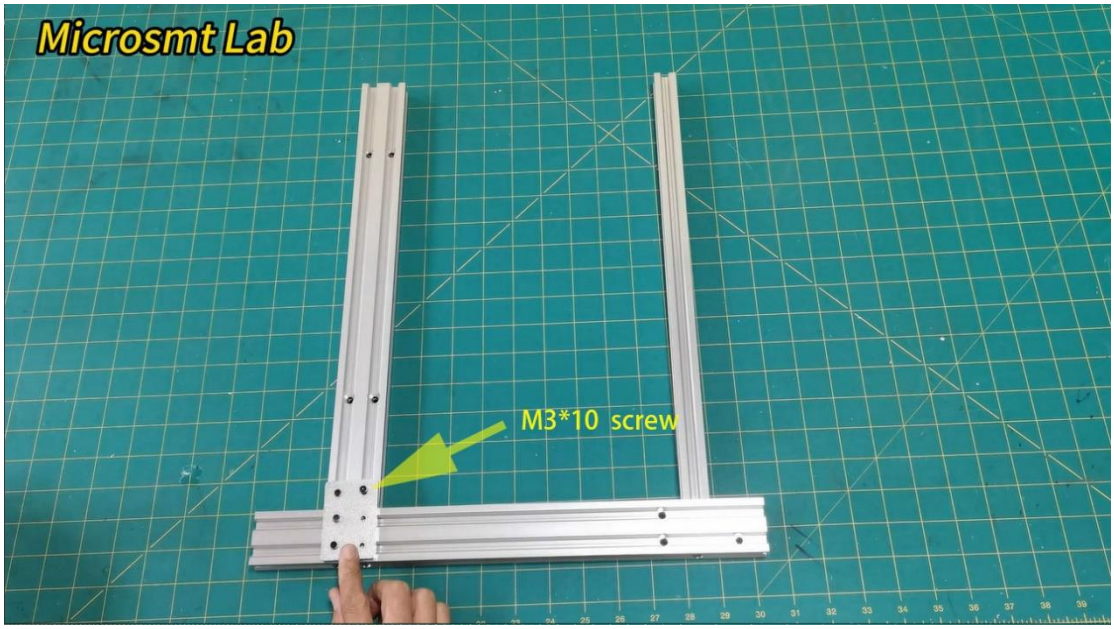
Step 1：Install 380mm 4020 aluminum profiles and 400mm 4020 aluminum profiles.



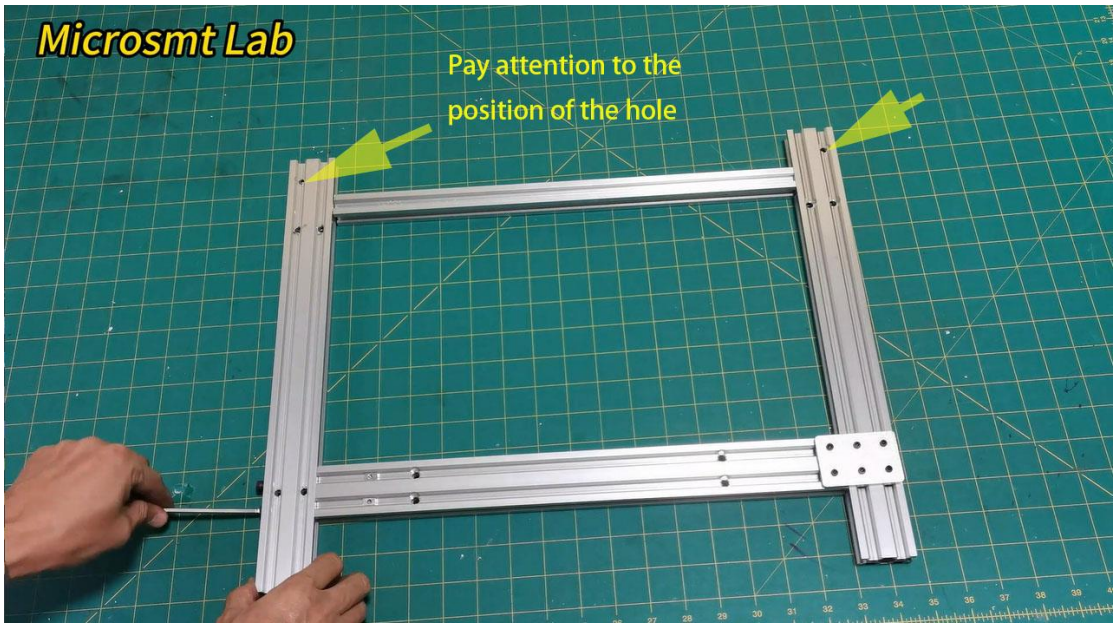
Step 2：Install 400mm 2020 aluminum profiles.

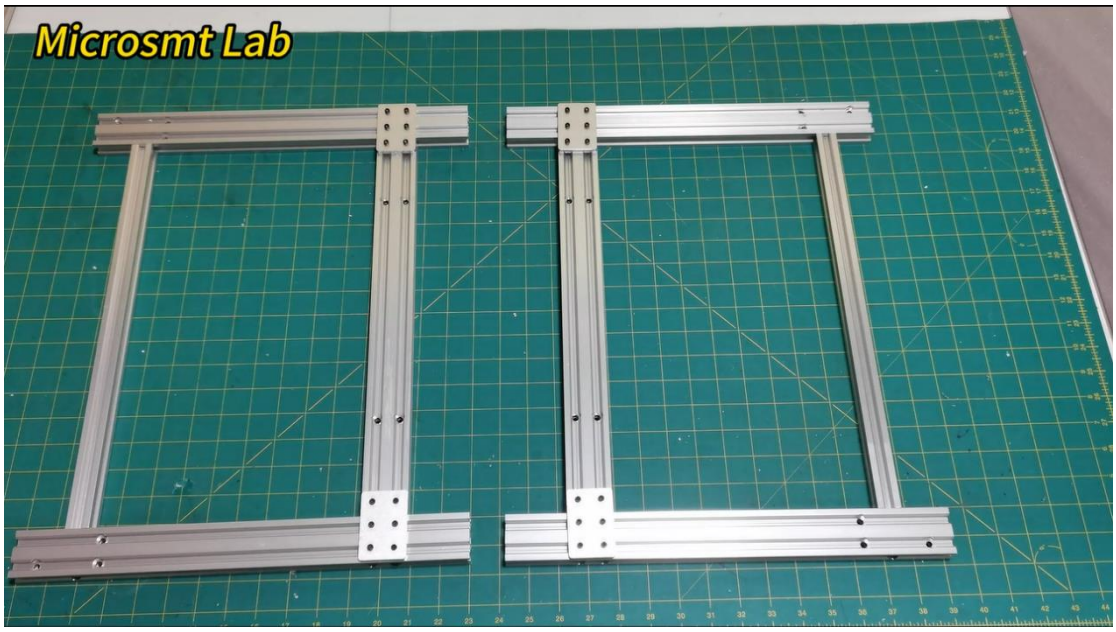


Step 3：Install B5 locking aluminum plate.

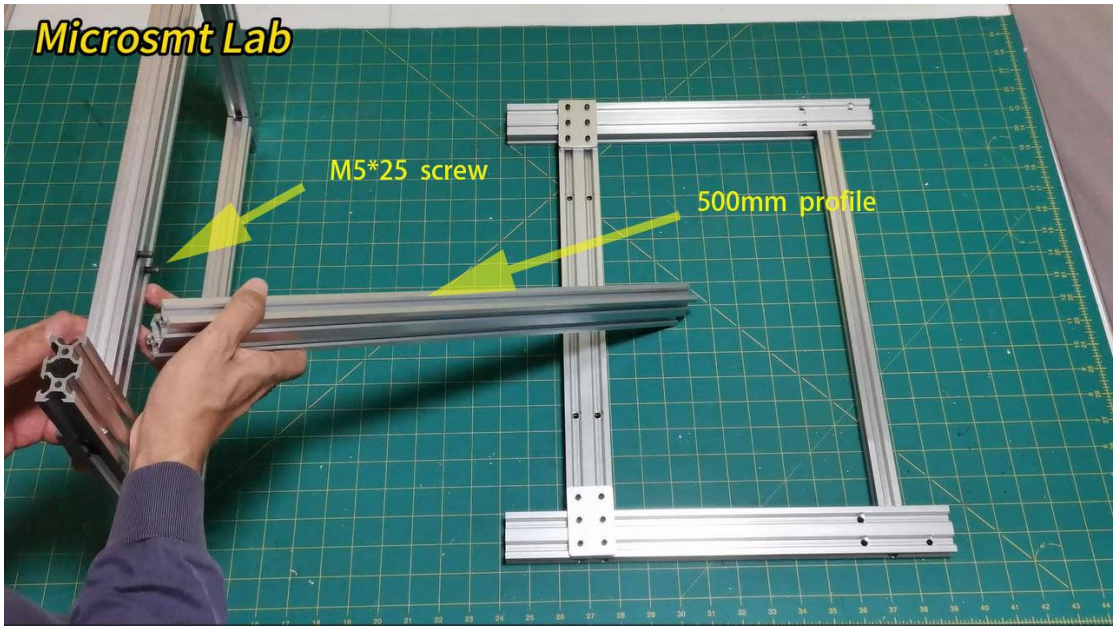


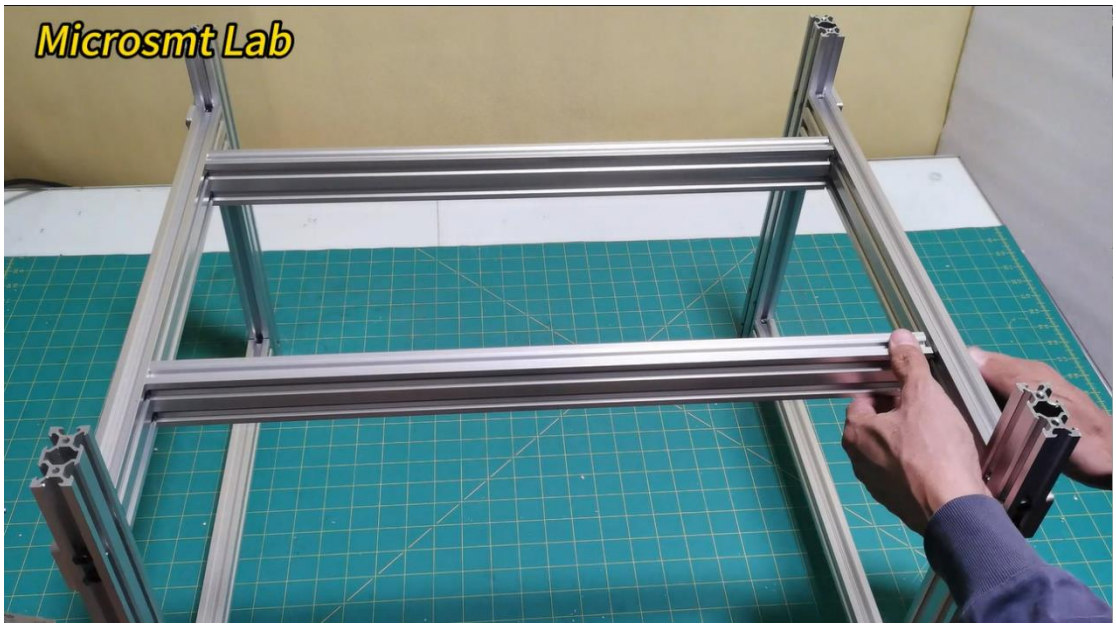
Step 4：



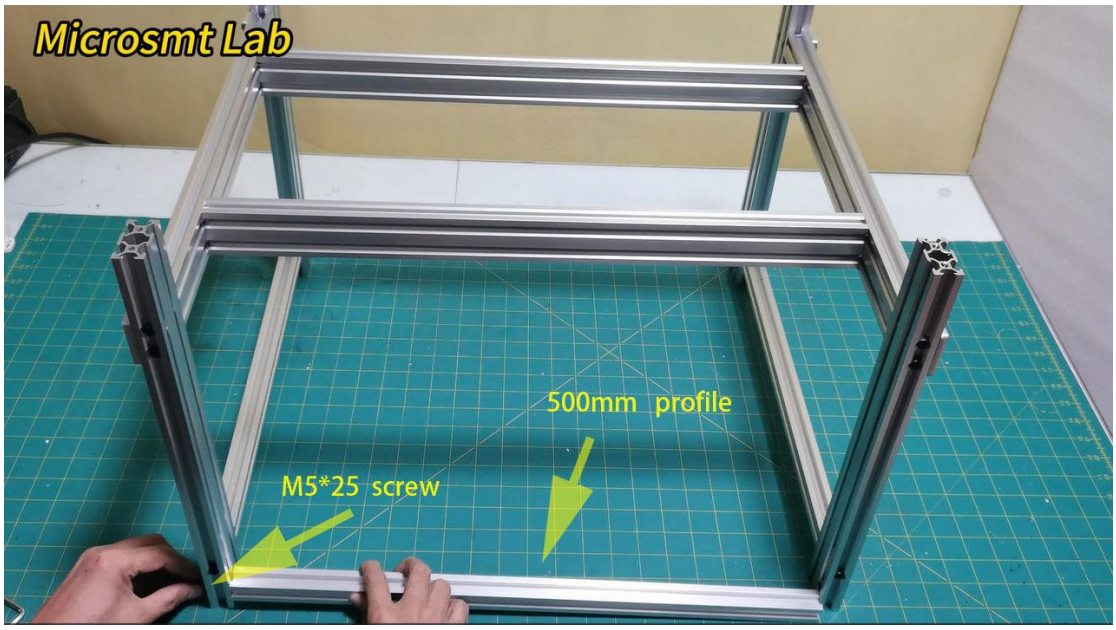


Step 5：Install 500mm 4020 aluminum profile.

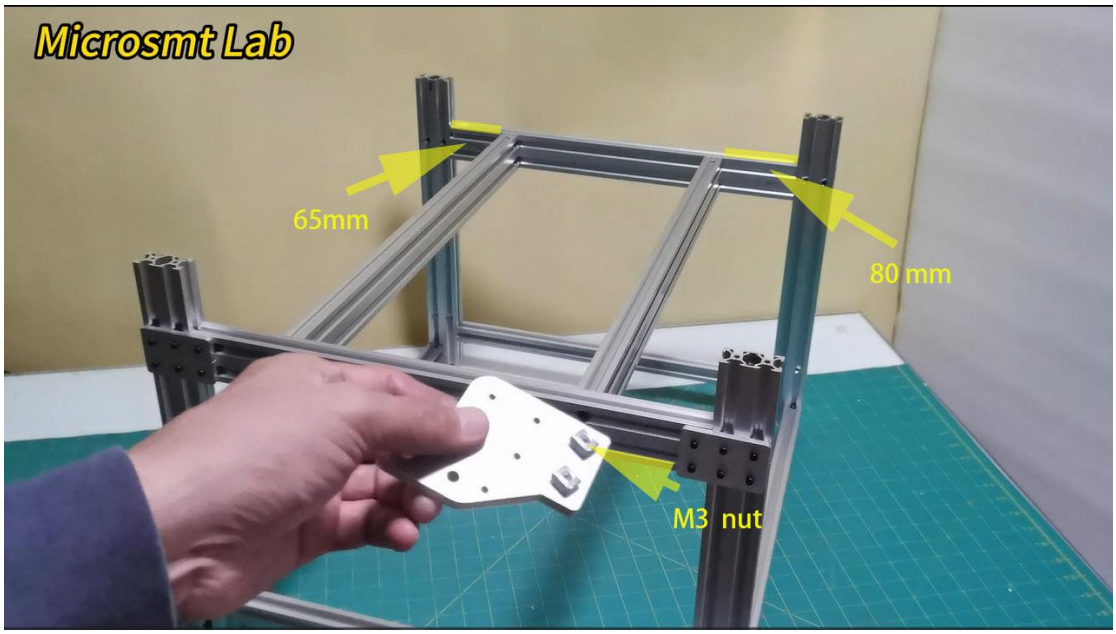


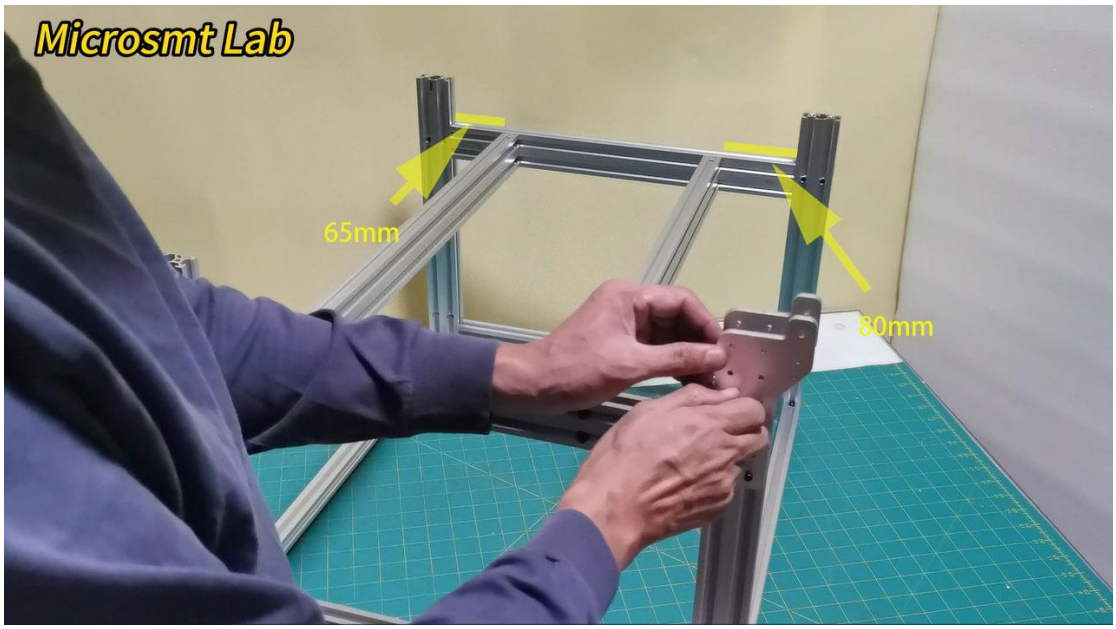


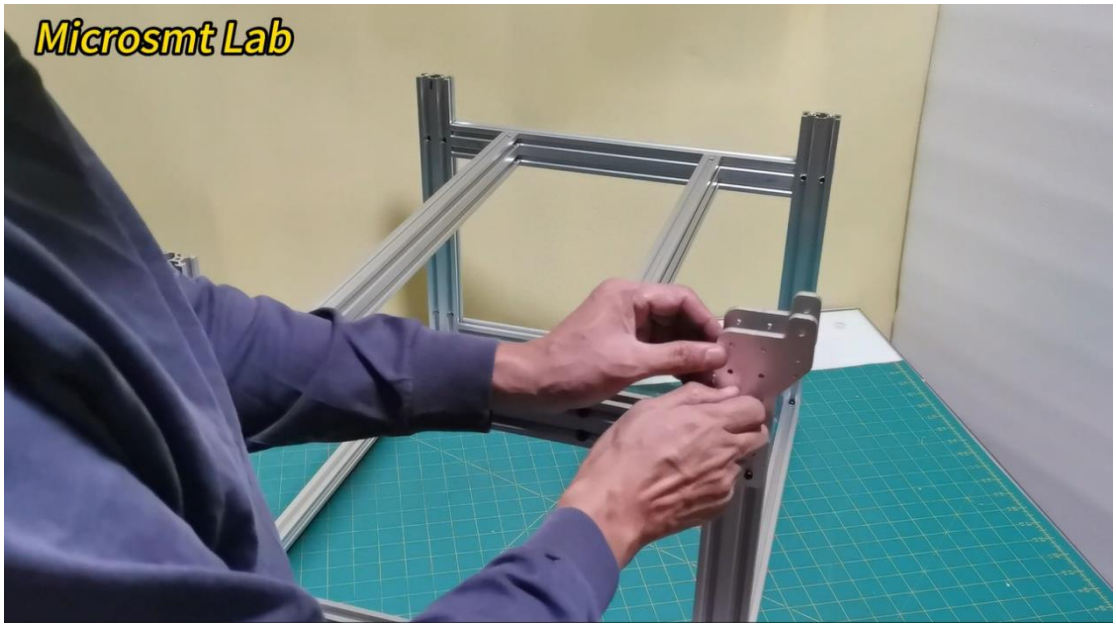
Step 6：Install 500mm 2020 aluminum profiles.



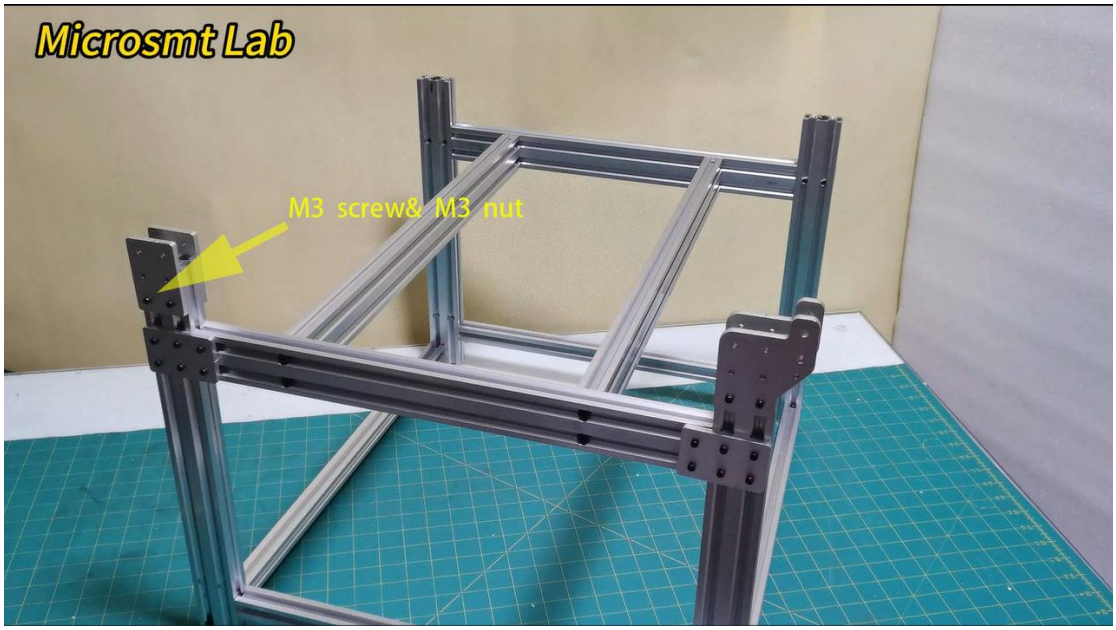
Step 7：Install B2 locking aluminum plate.



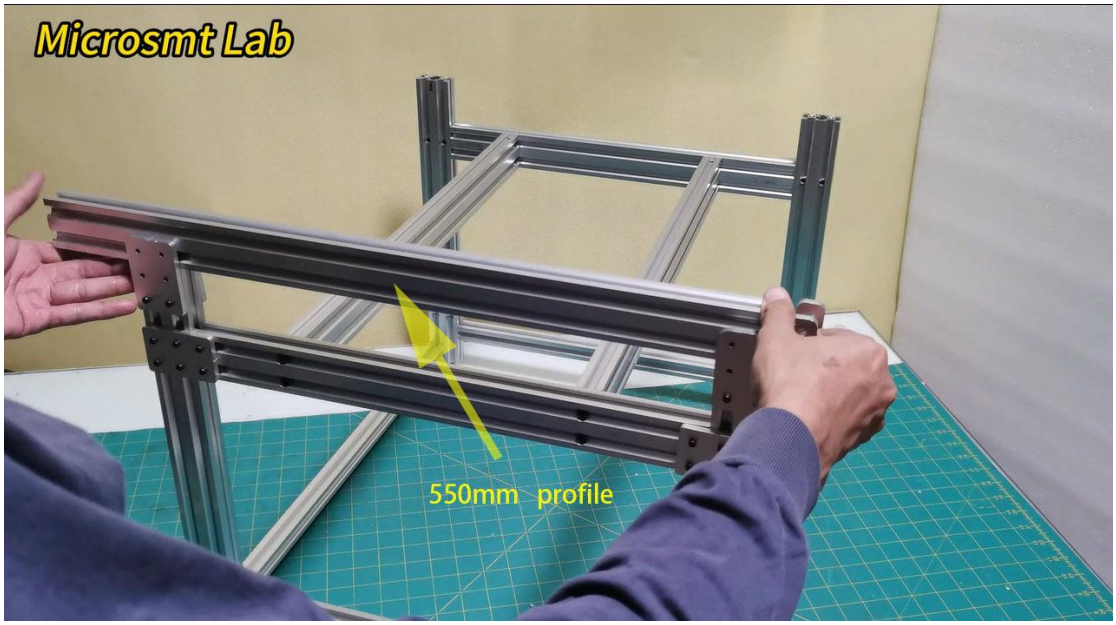


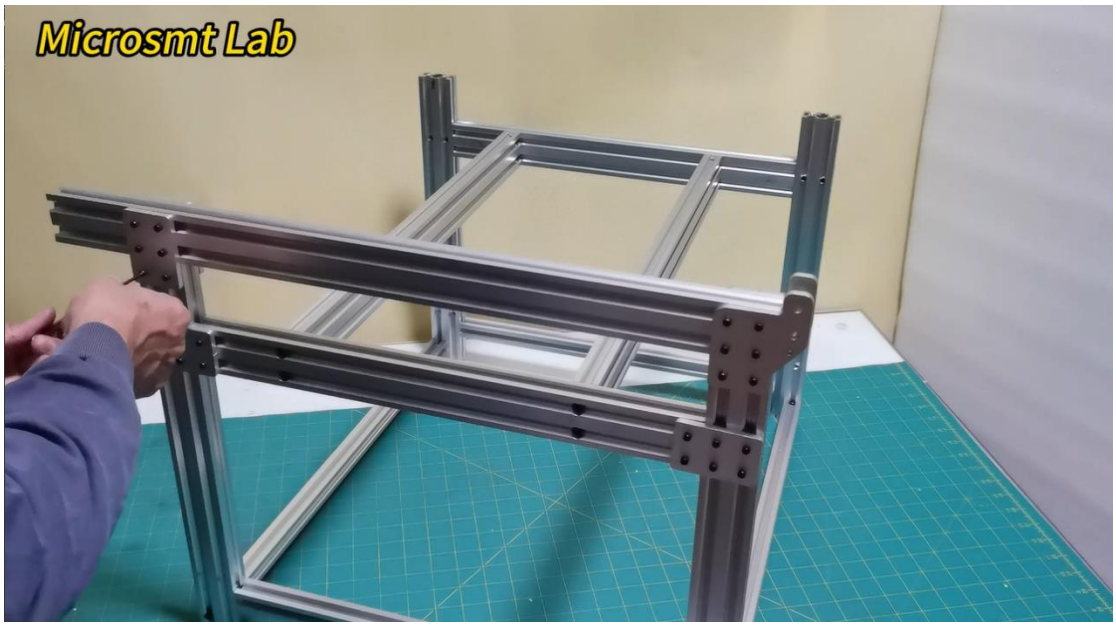


Step 8：Install B5 locking aluminum plate.

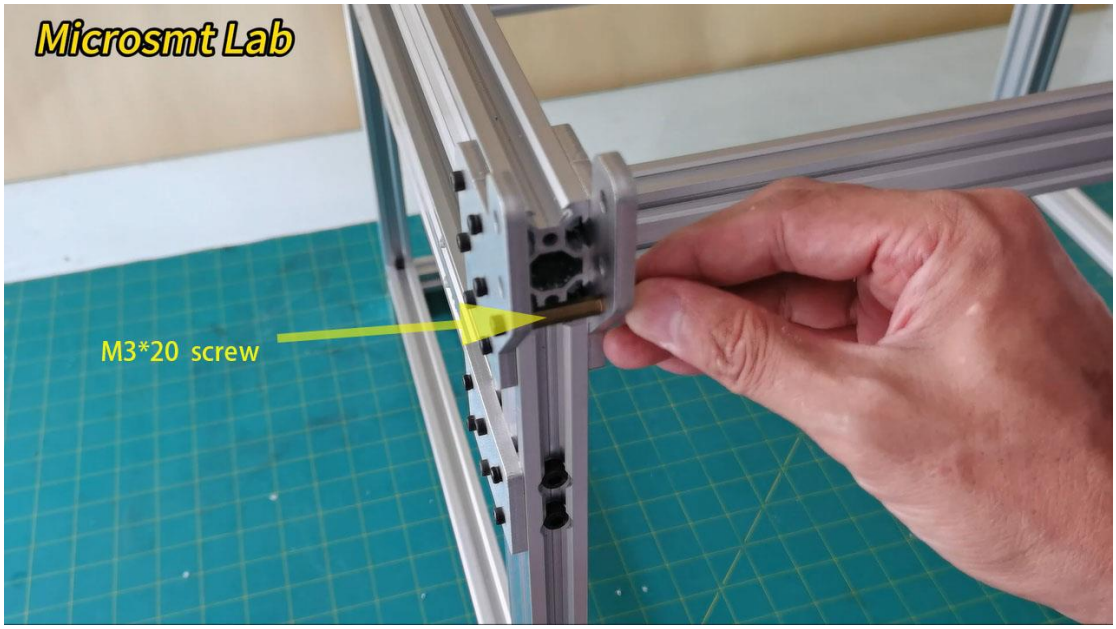


Step 9： Install 550mm 4020 aluminum profile.

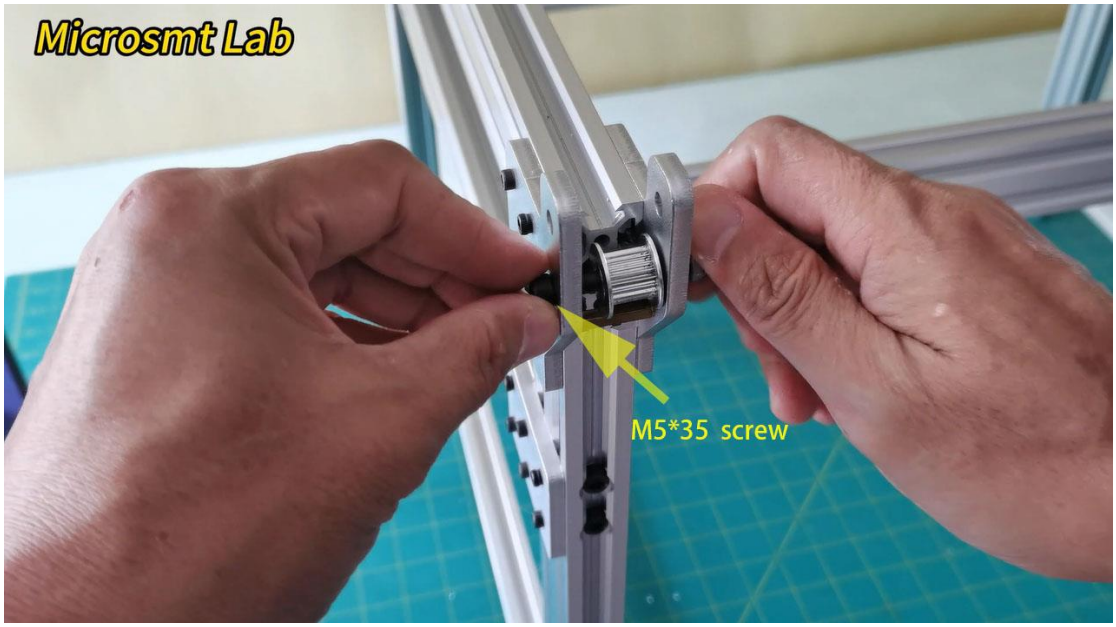


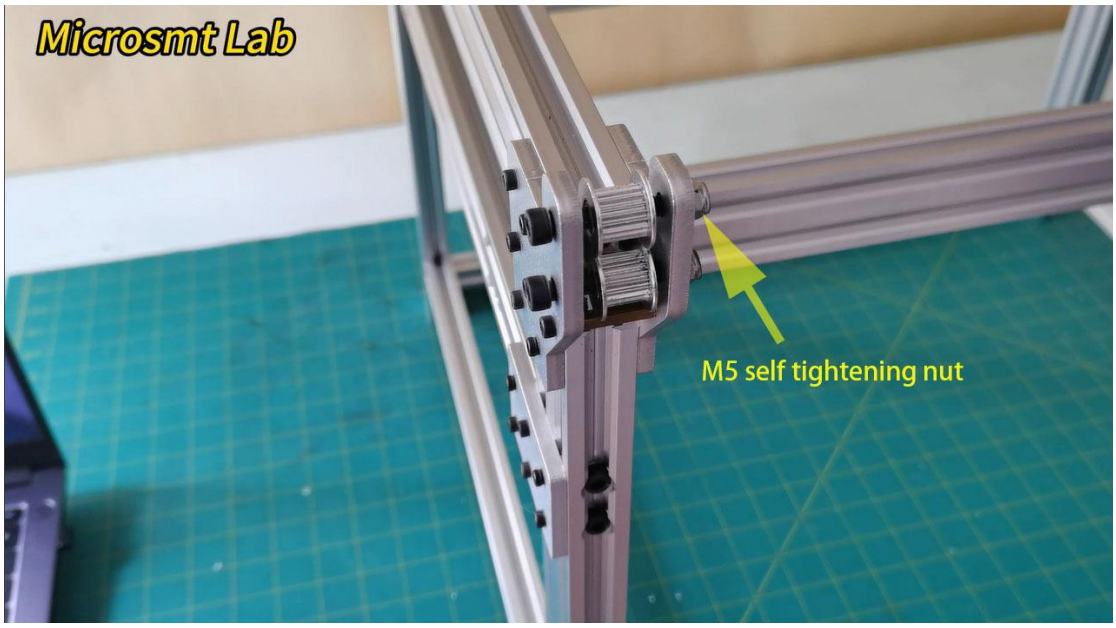


Step 10：Install M3 * 20 copper nut.



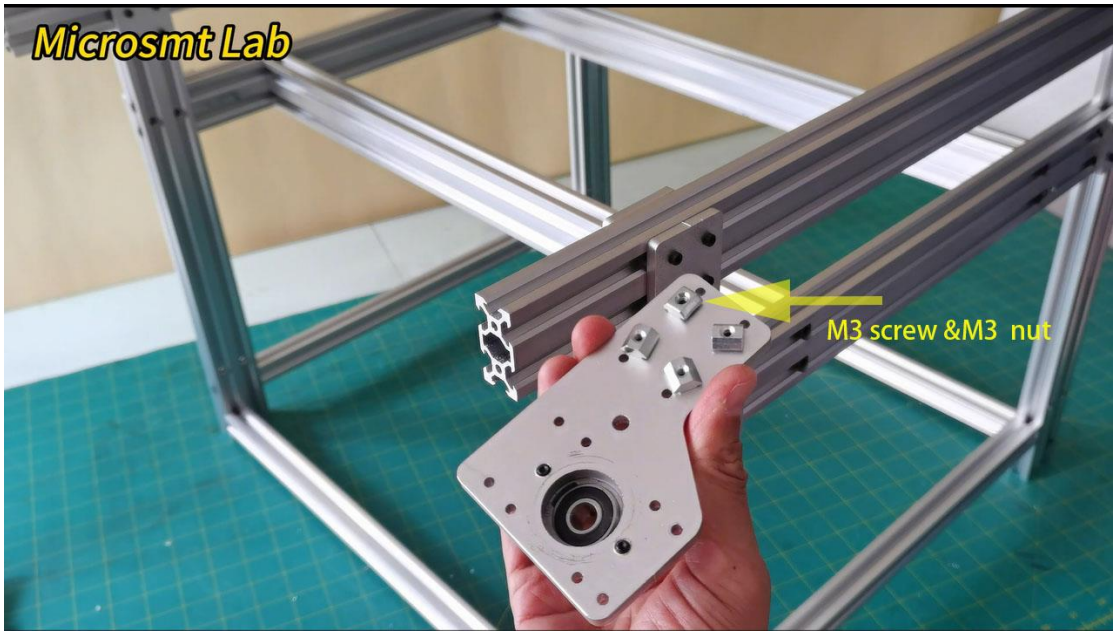
Step 11：Install 20T　synchronous wheel.

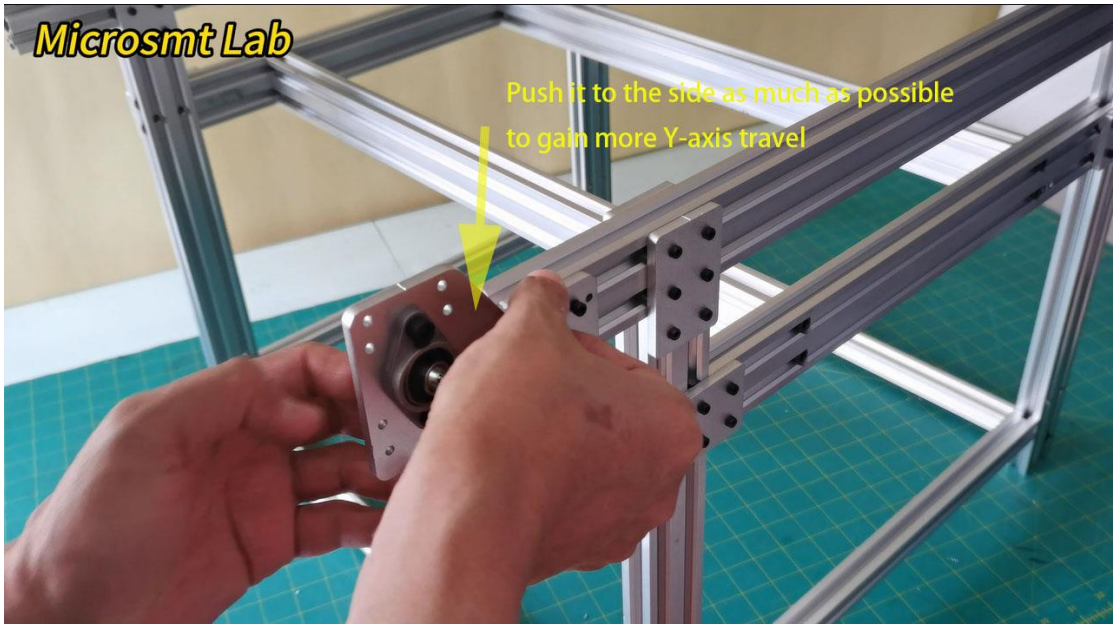



Microsmt Lab
M5 self tightening nut

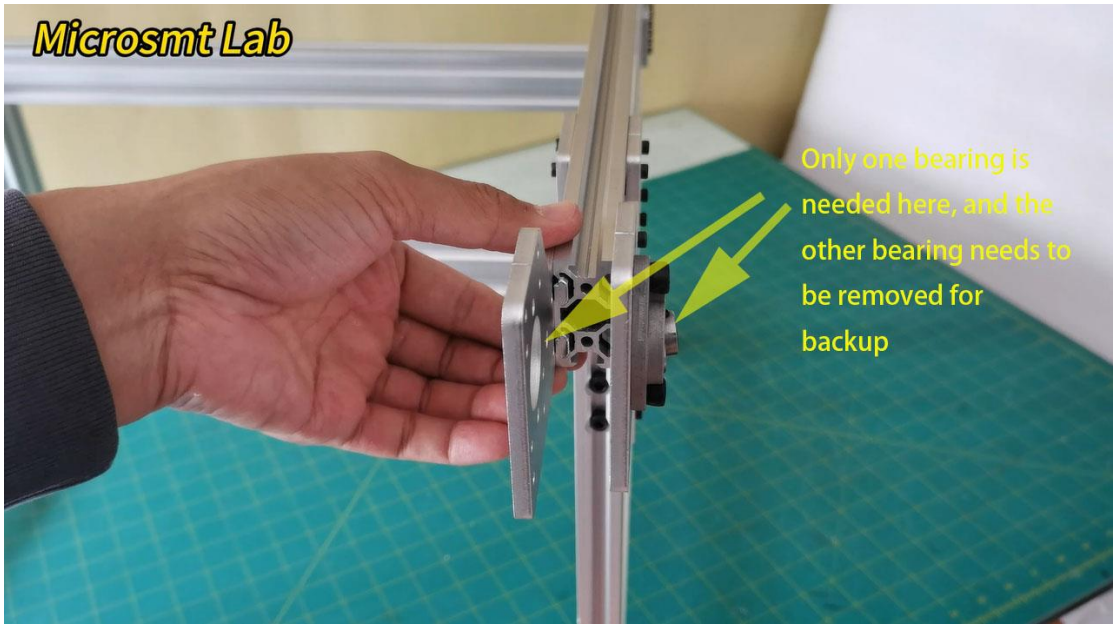

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Step 12：Install B3 locking aluminum plate (left side).

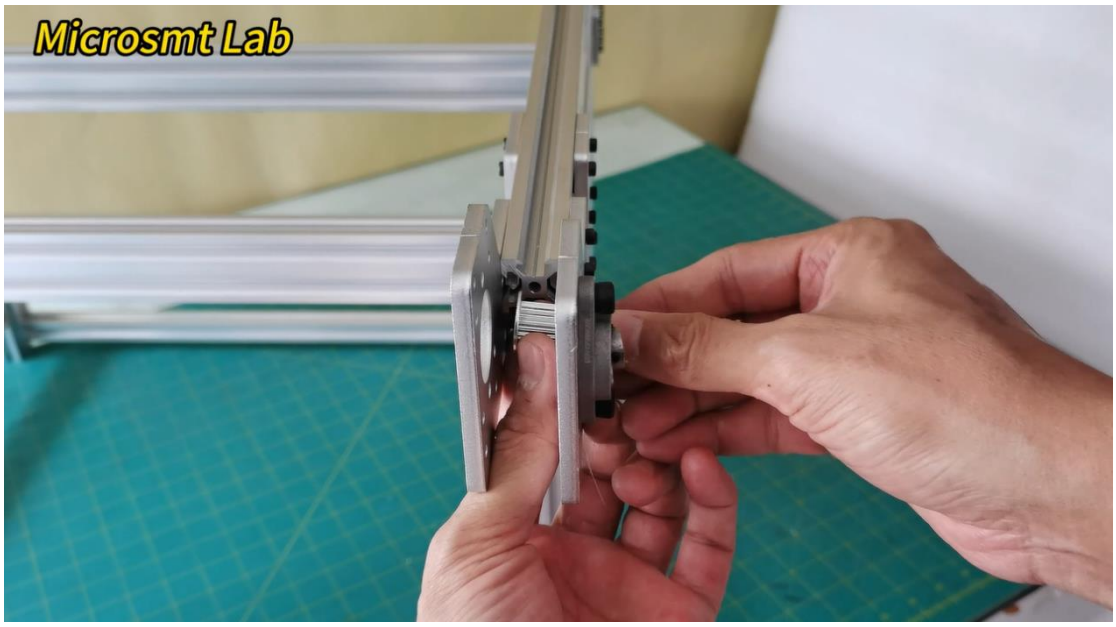




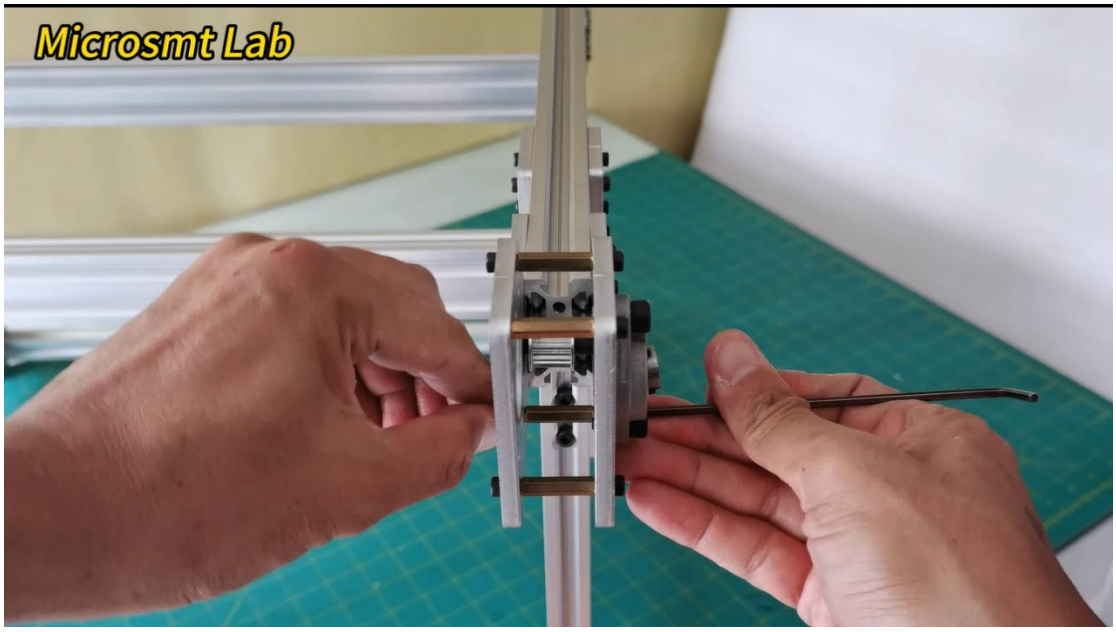
Step 13：



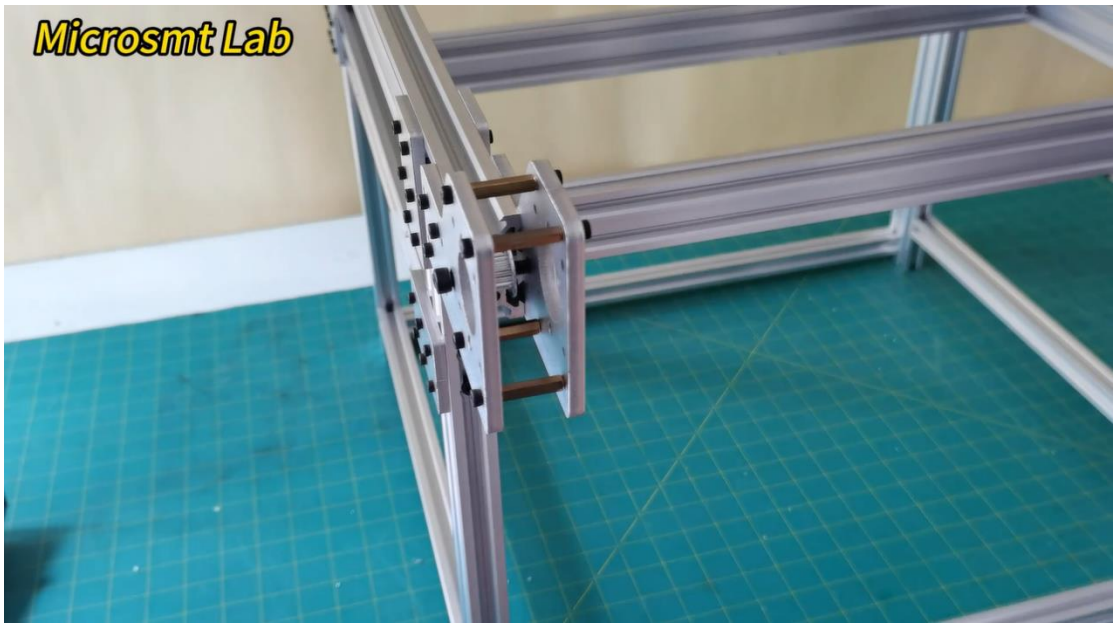
Step 14：Install 20T synchronous wheel.



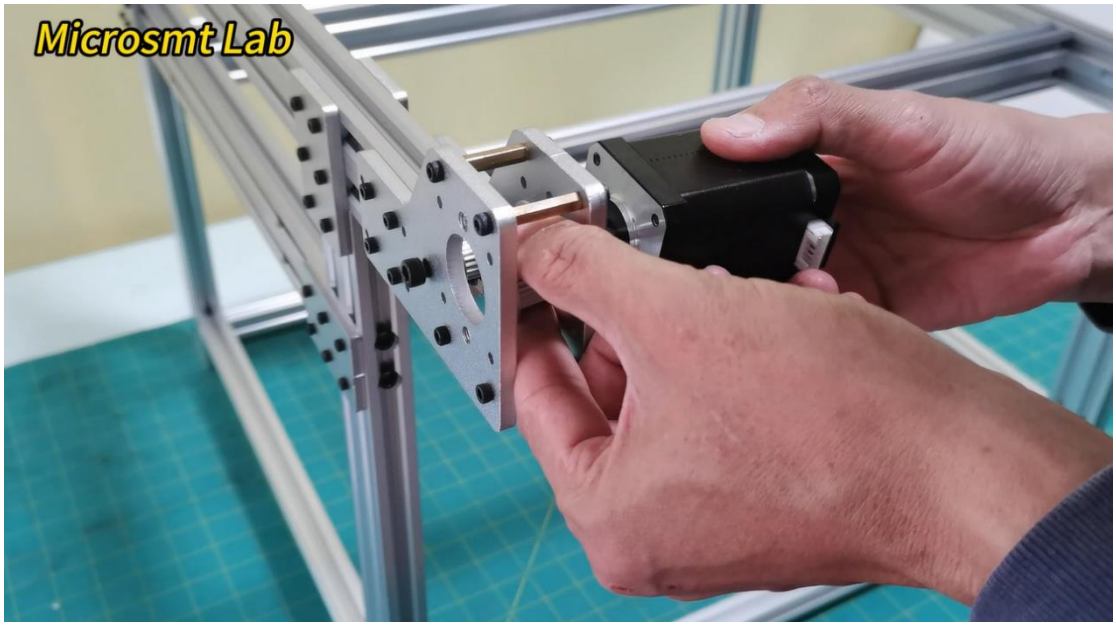
Step 15：Install M3 * 20 copper nut.

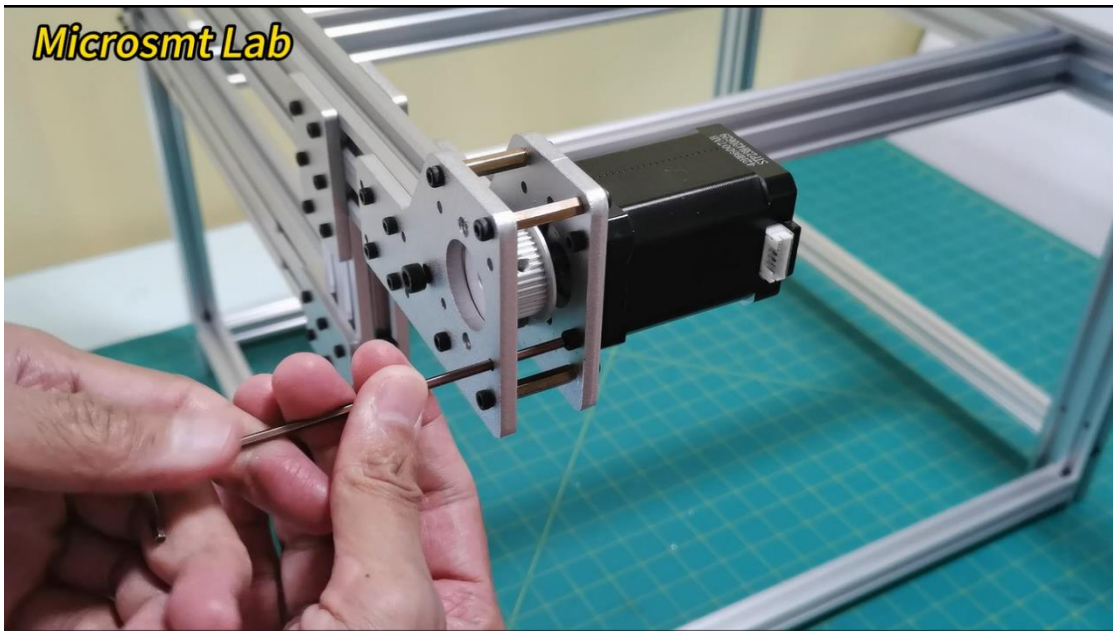


Step 16：Install B3 locking aluminum plate (right side).

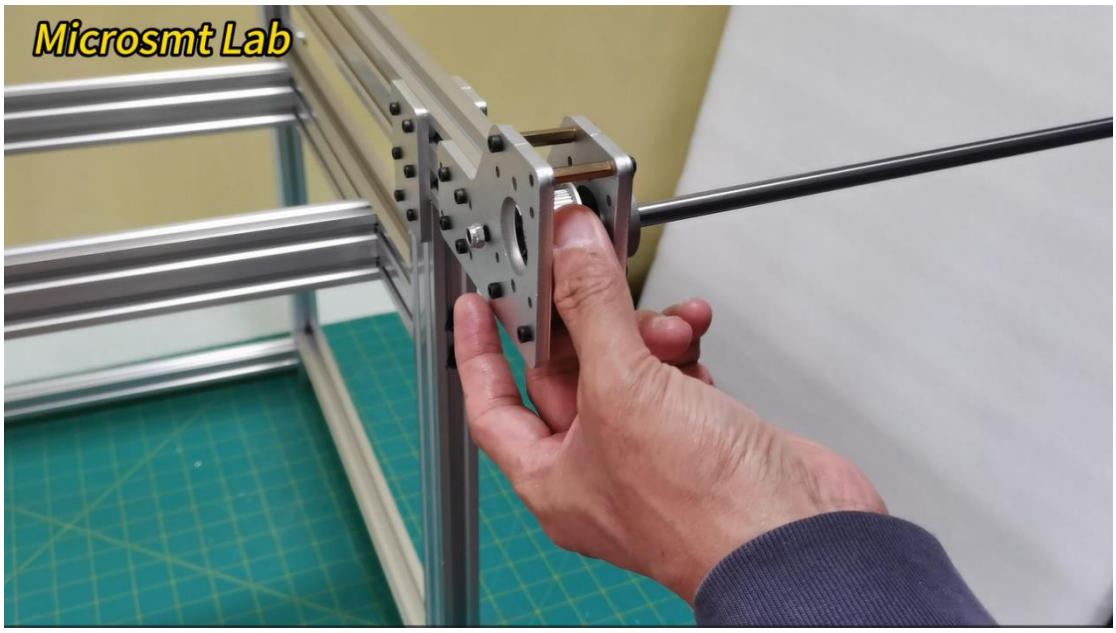


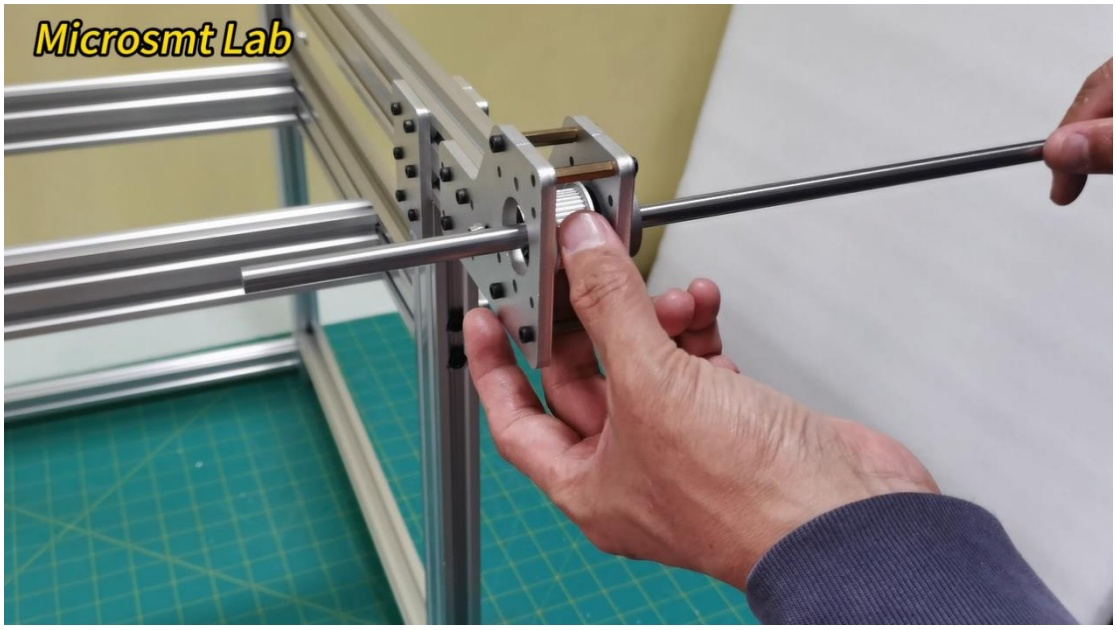
Step 17：Install dual axis motor and 40T synchronous wheel(right).



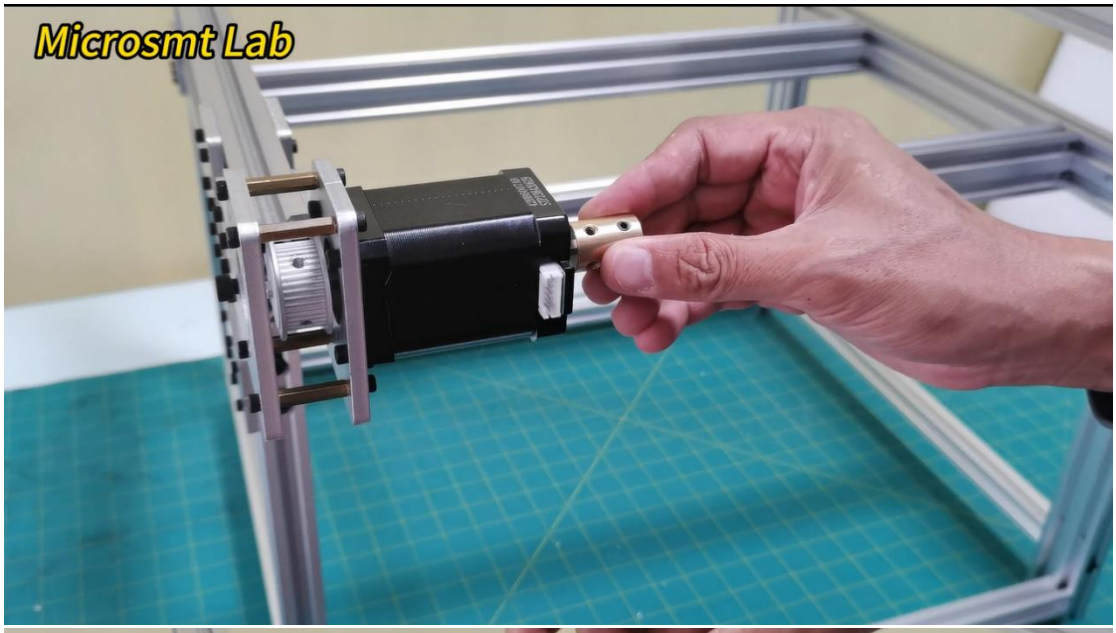


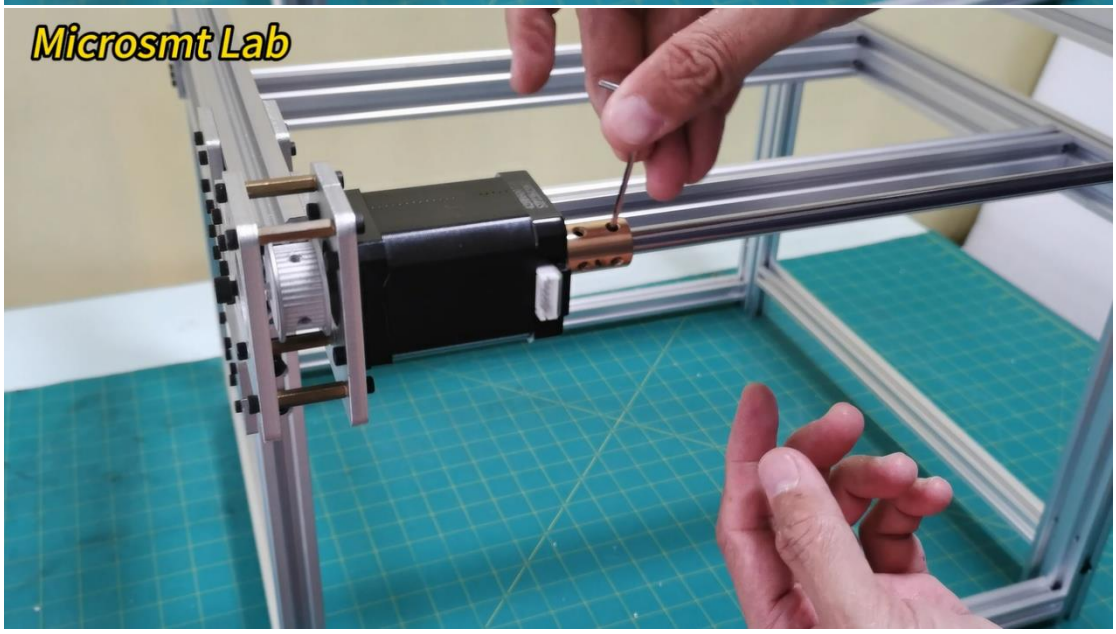
Step 18：Install 8mm synchronous shaft and 40T synchronous wheel（left）.

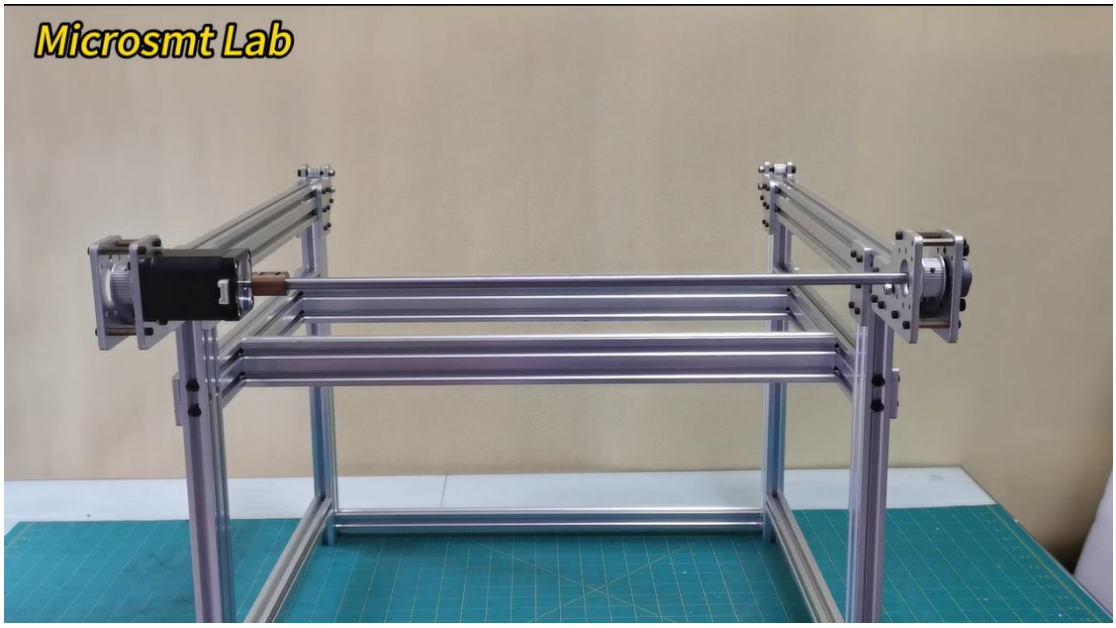




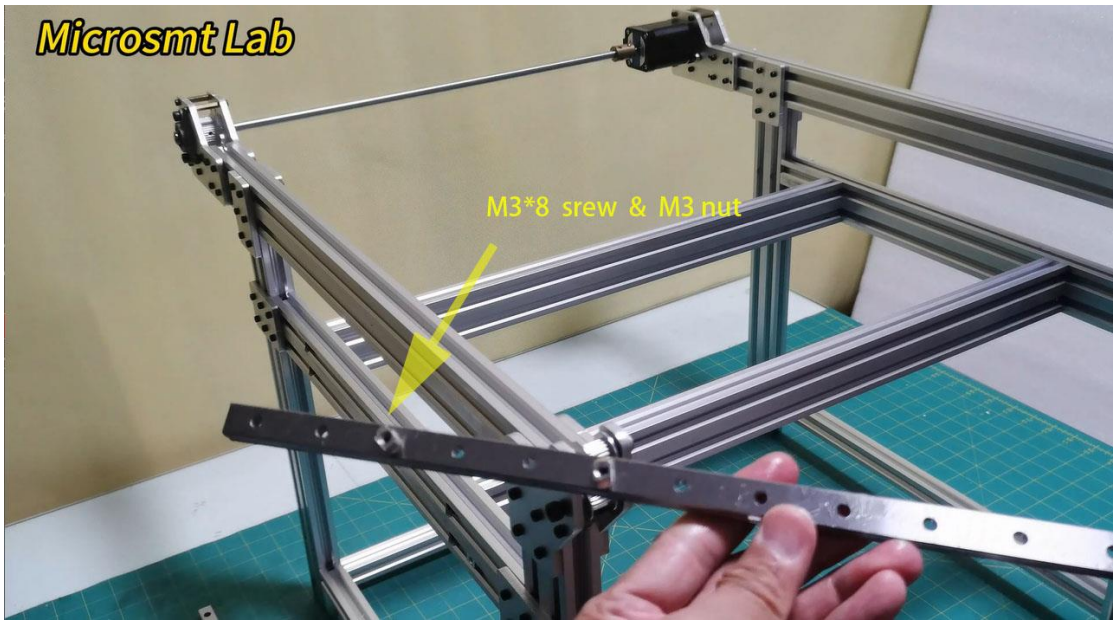
Step 19：Install the coupling and lock the synchronous shaft by M4 scew.

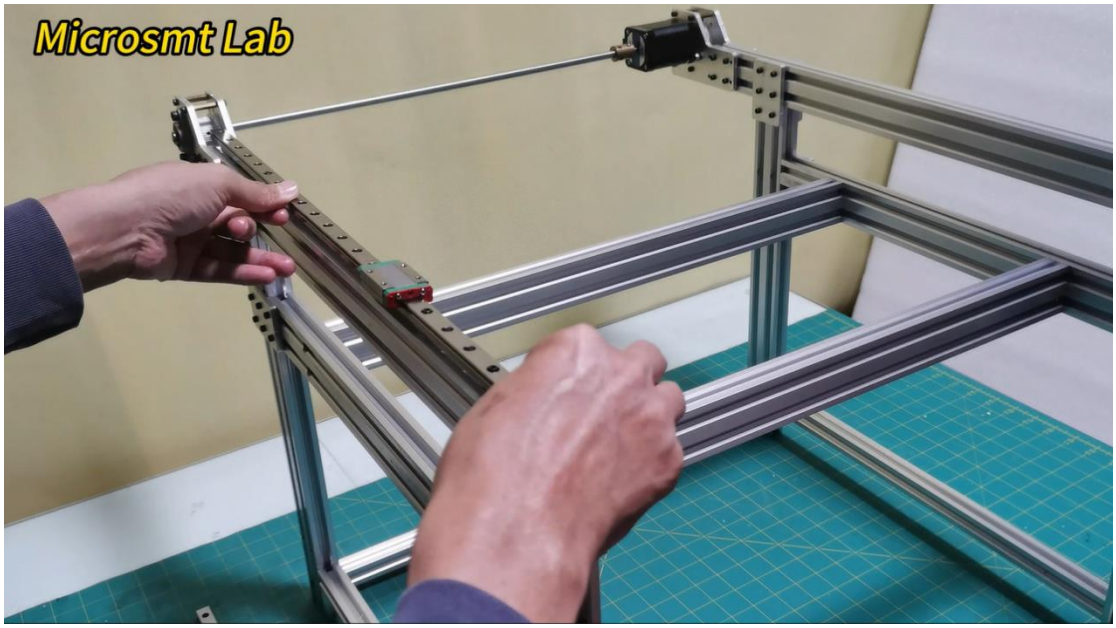




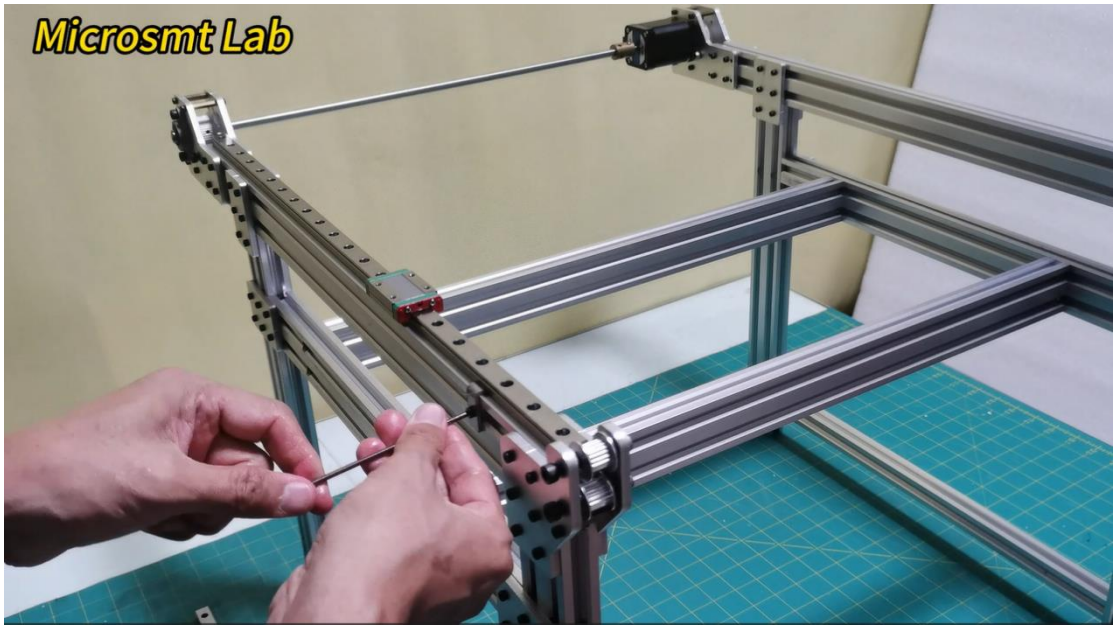


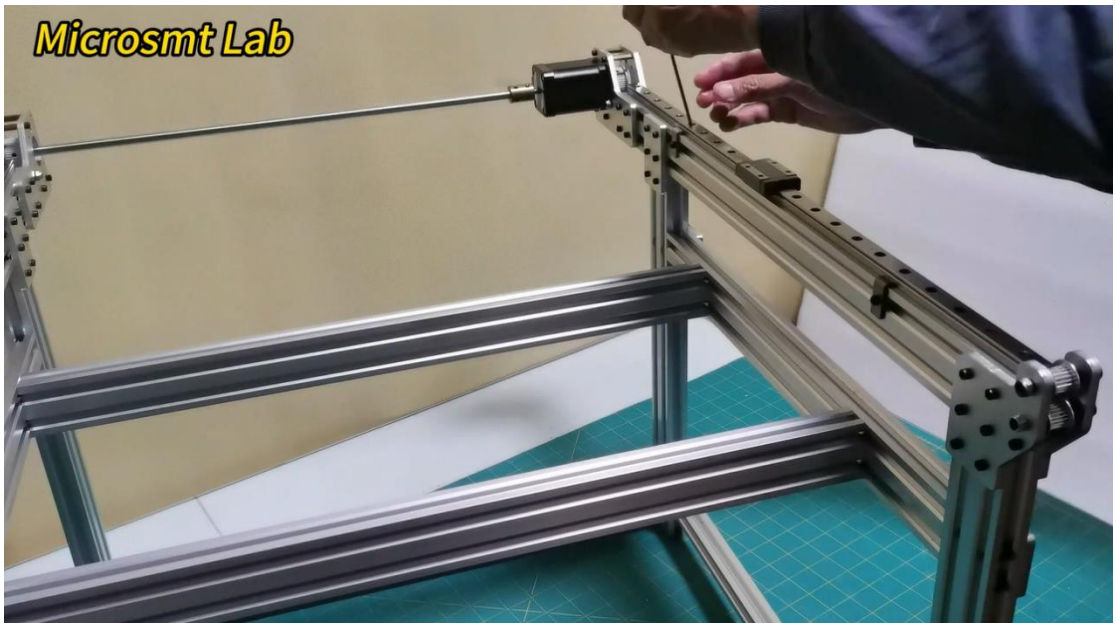
Step 20：Install Y linear guide rail, Use circular nuts for easy replacement, or you can also use square nuts. The quantity is 5-6 pieces, there is no need to put screws in every hole.



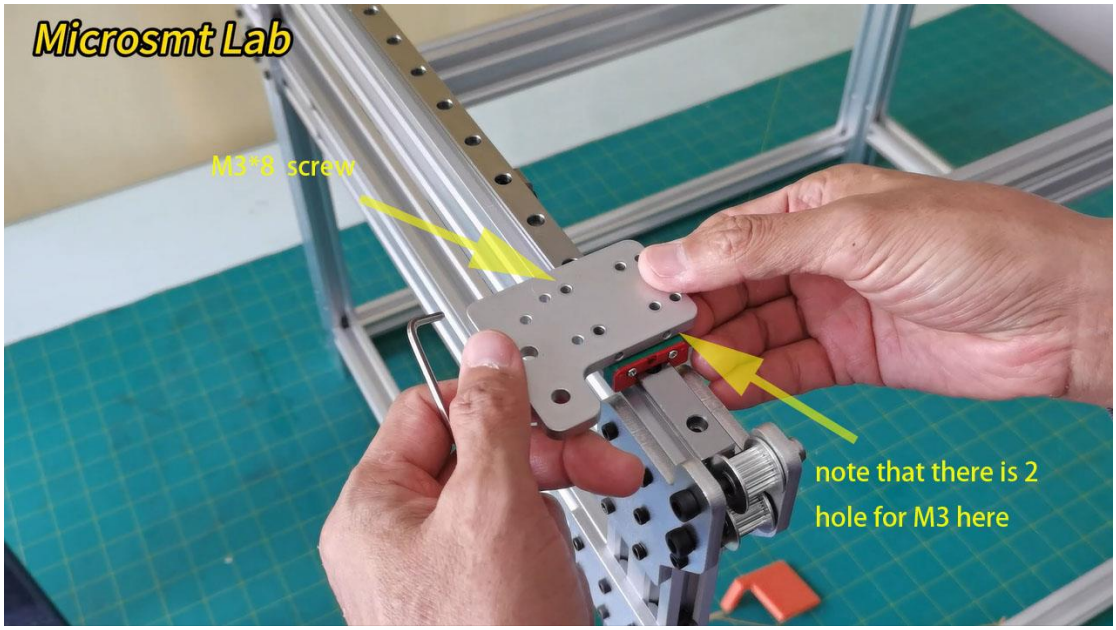


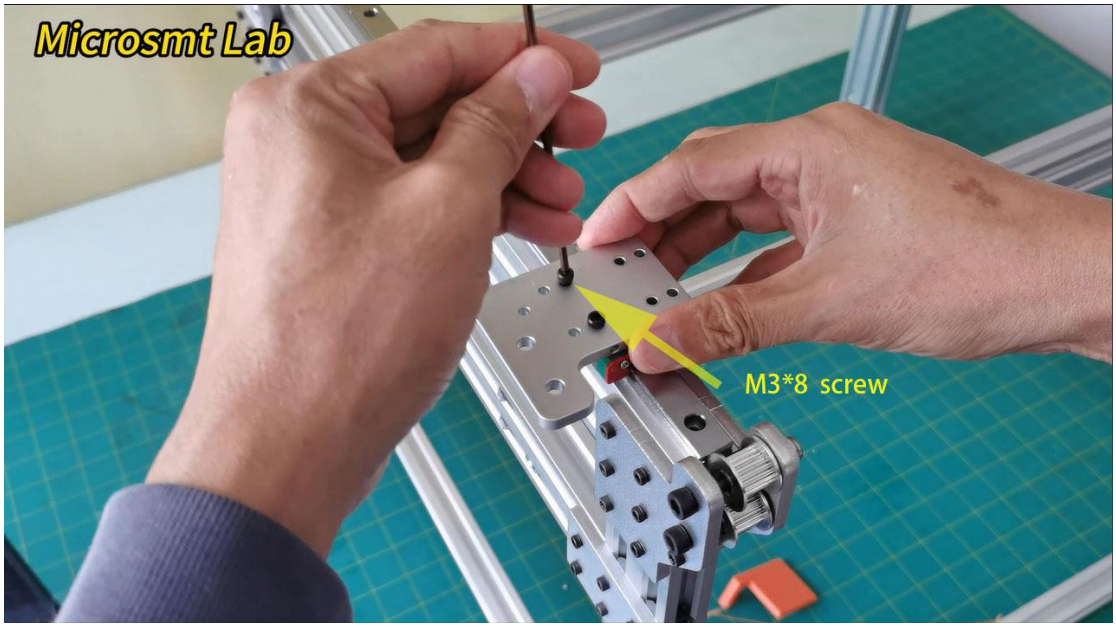
Lock the linear guide rail with 4PCS limit steel plates.



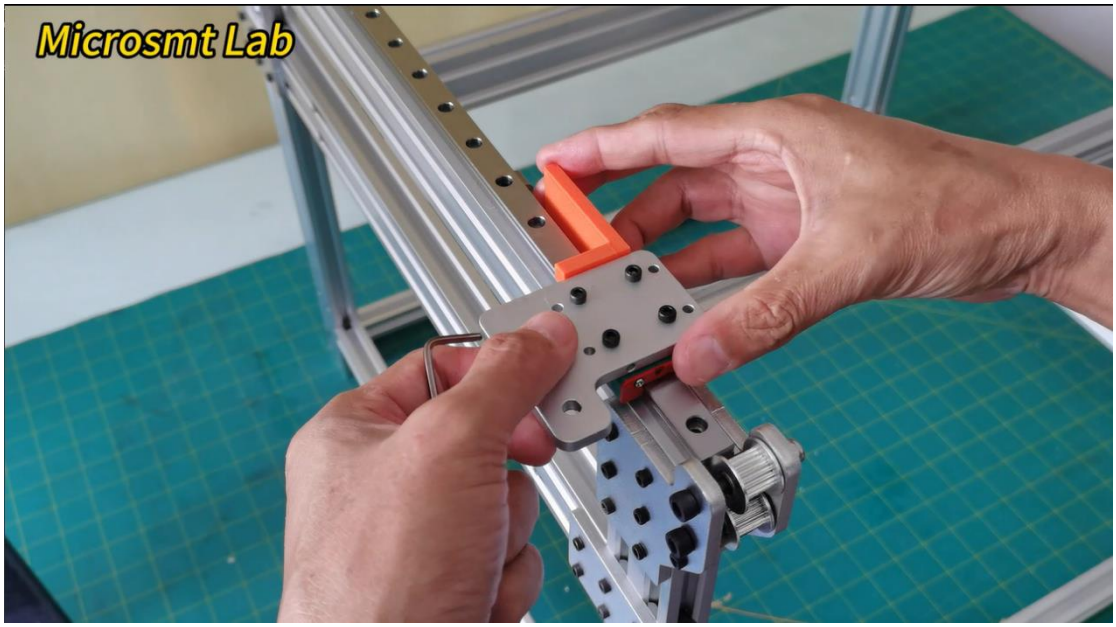


Step 21：Install B4 aluminum plate(belt locked).

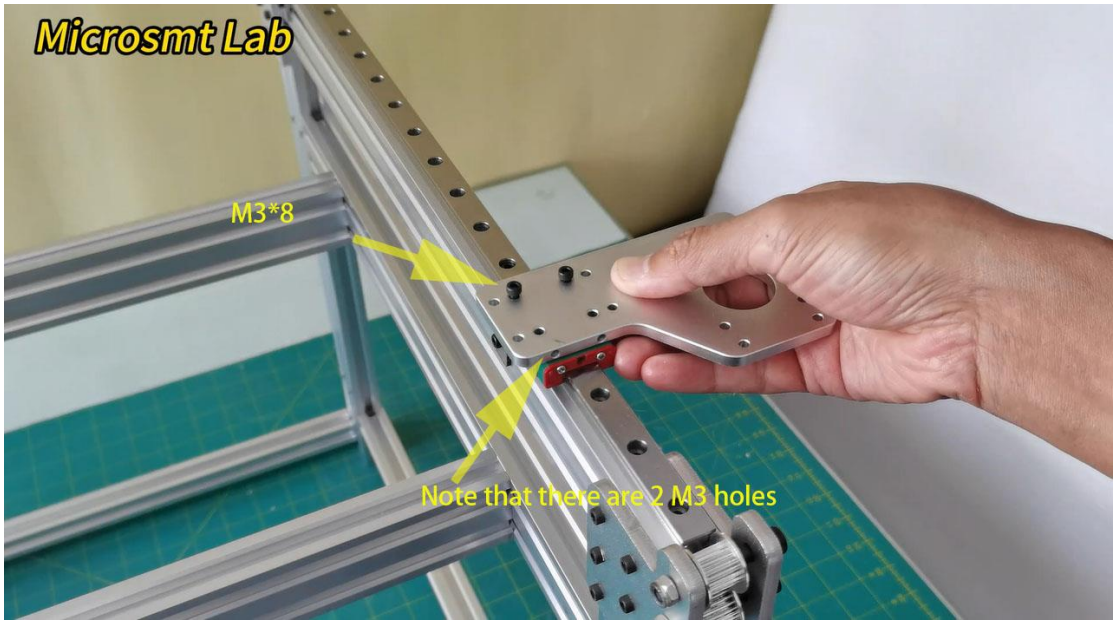




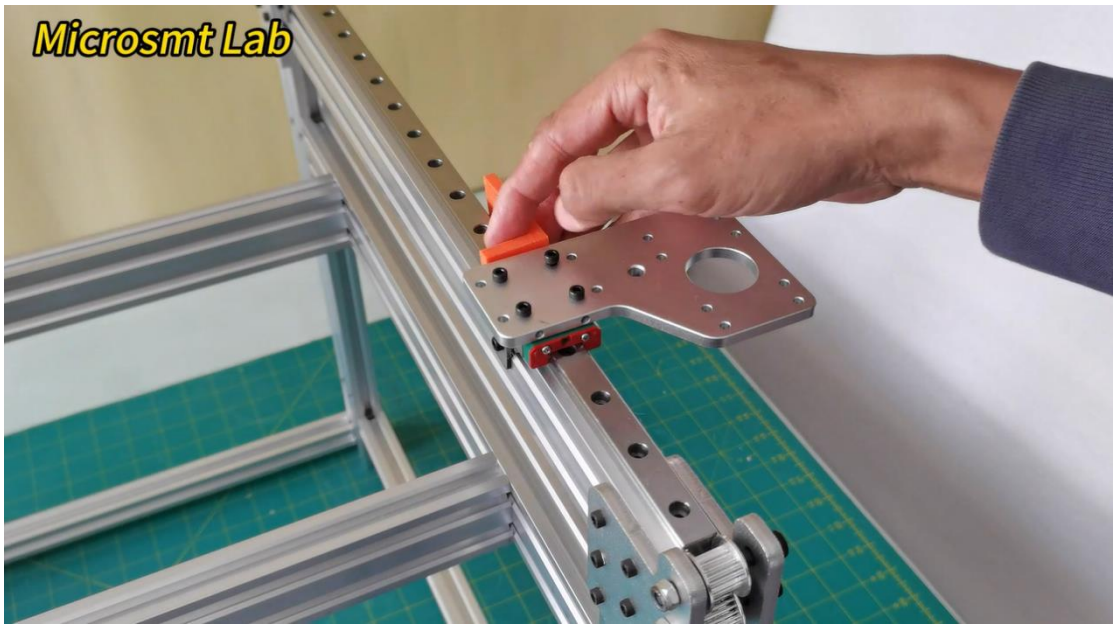
Make it perpendicular to the Y-axis .



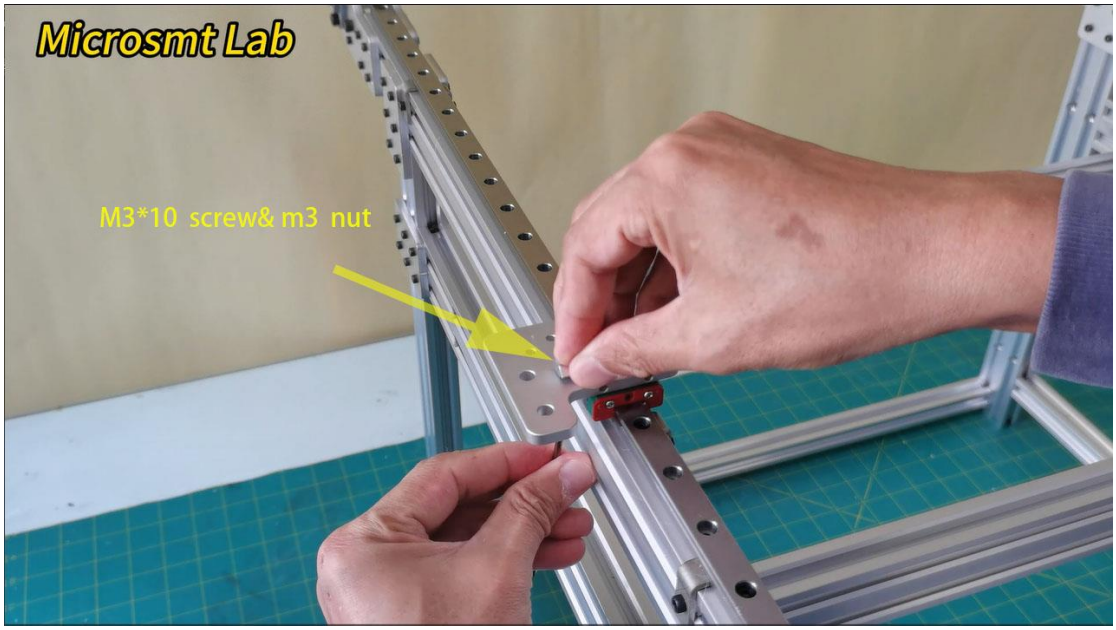
Step 22：Install B3 aluminum plate.

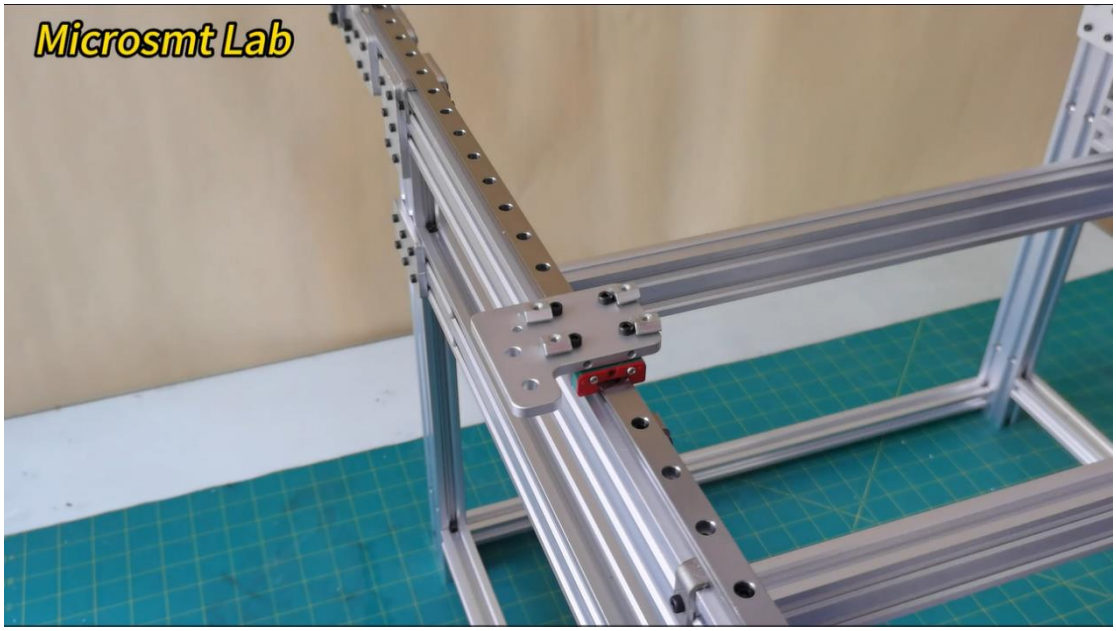


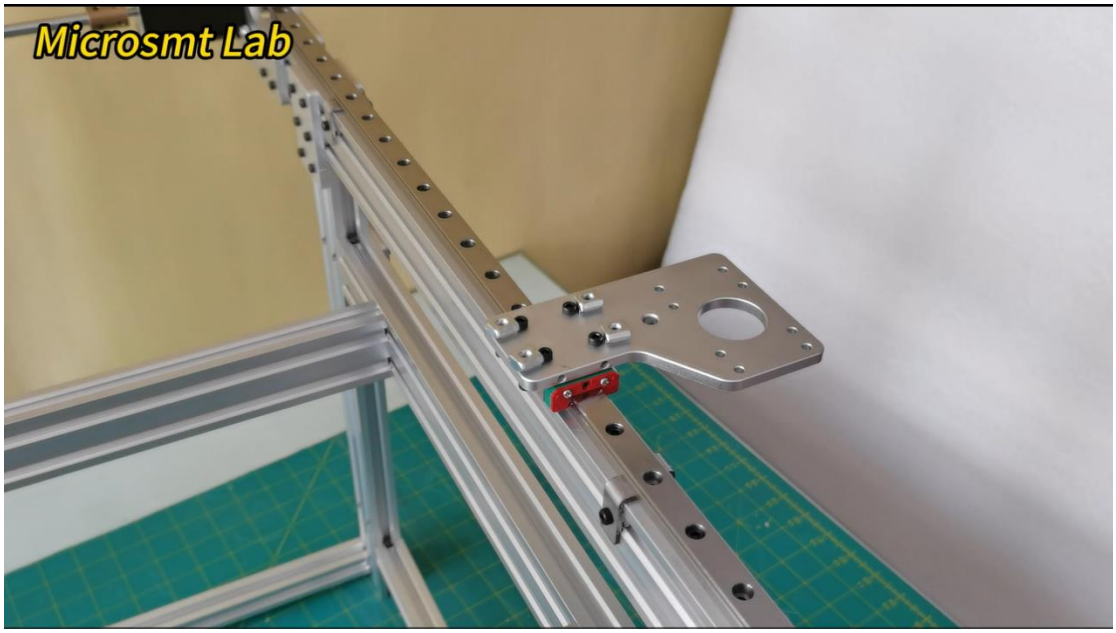
Make it perpendicular to the Y-axis.

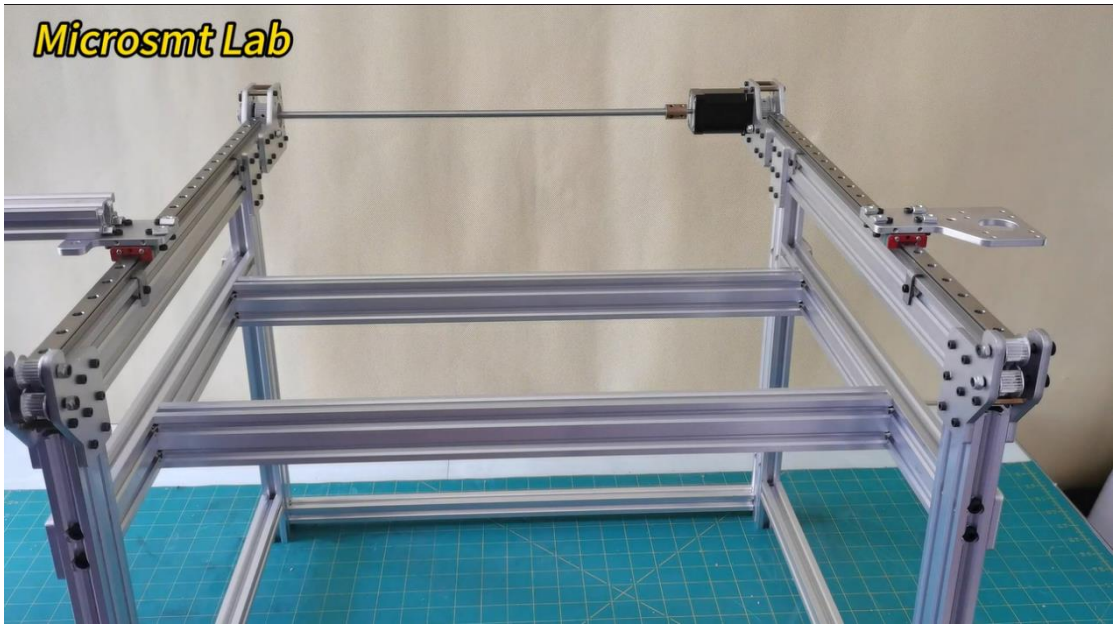


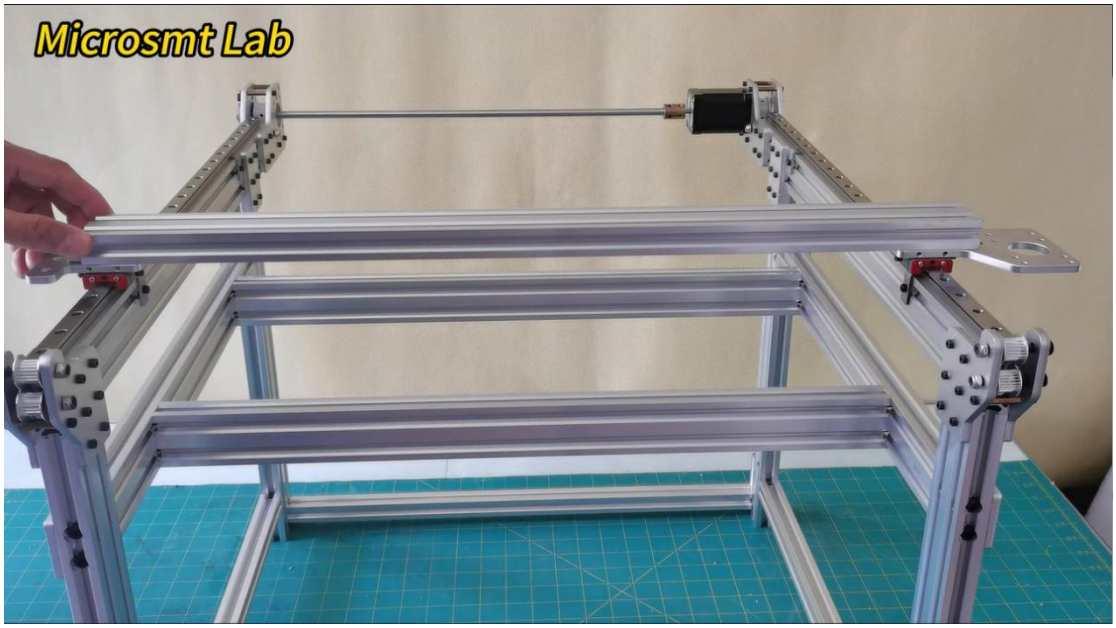
Step 23: Install 563mm 4020 aluminum profile.

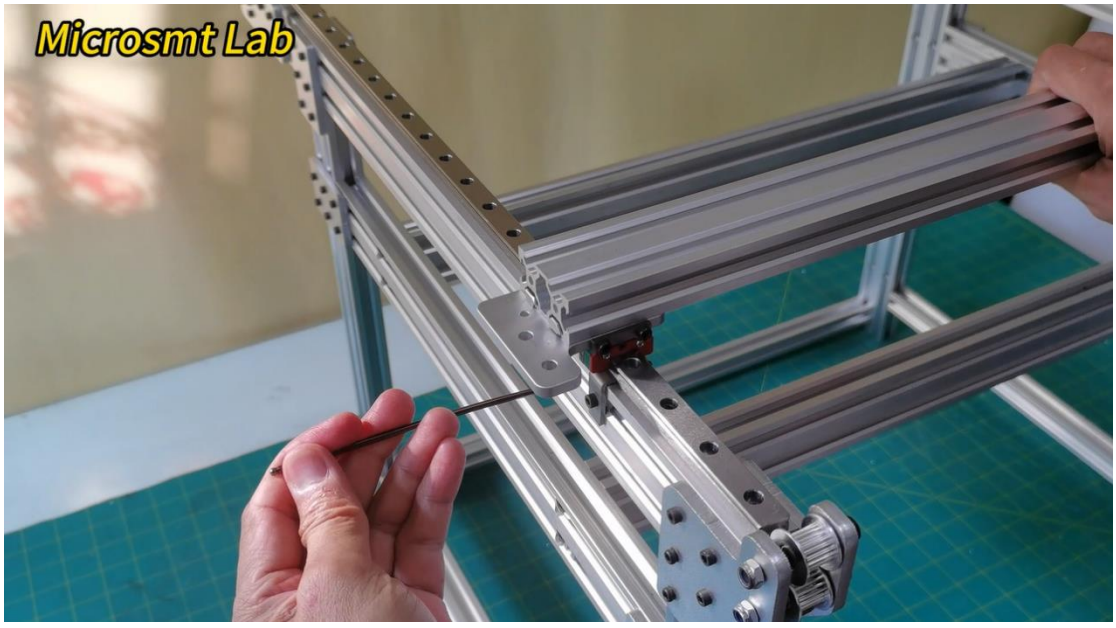




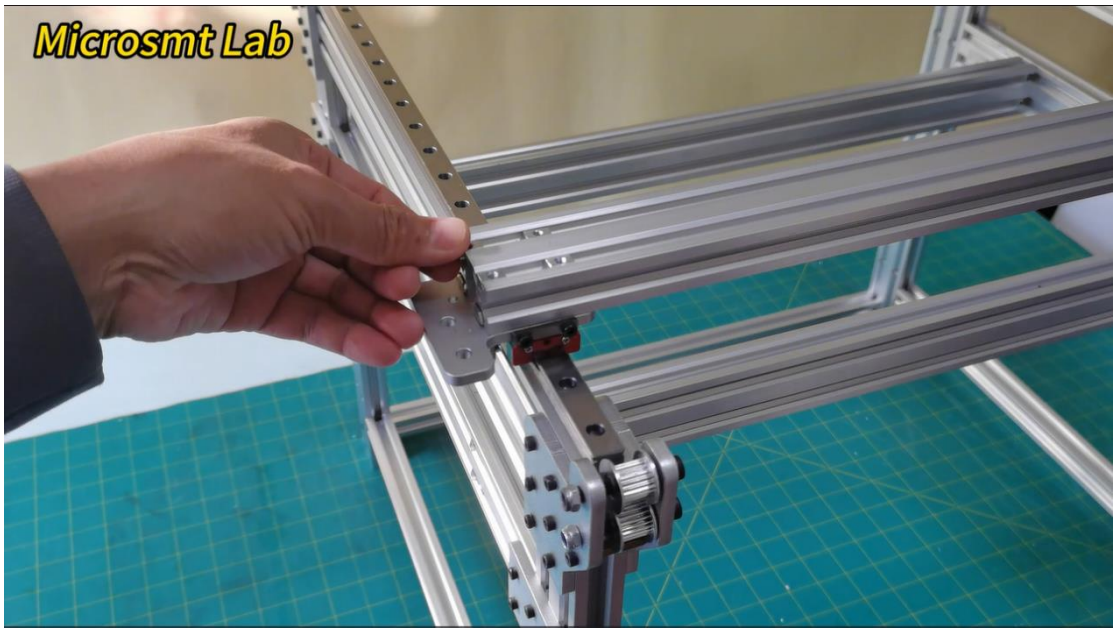

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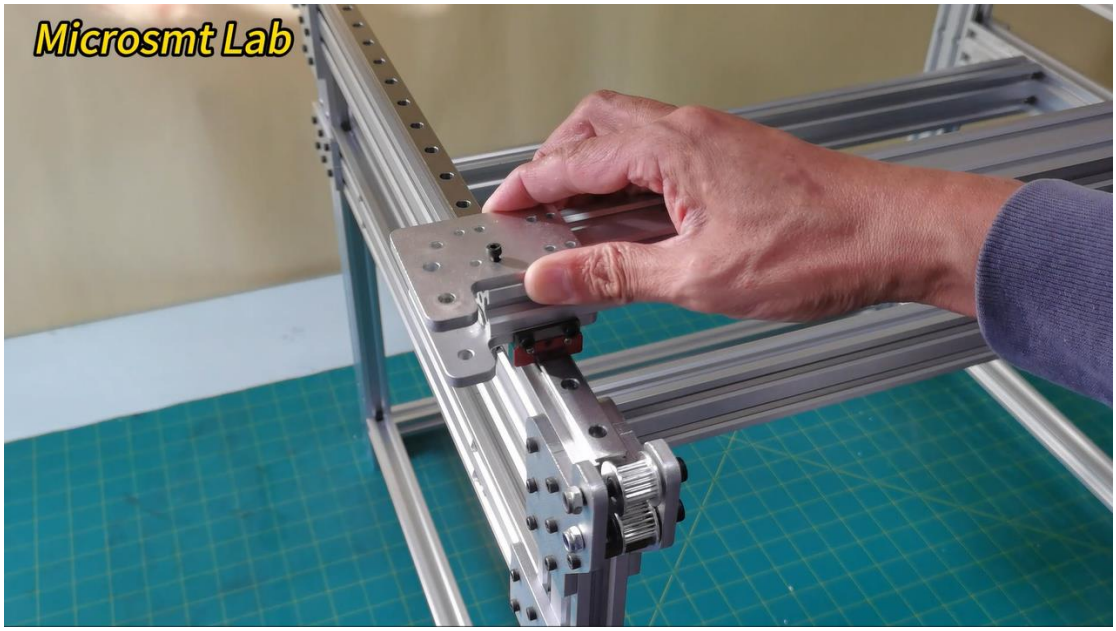

Microsmt Lab

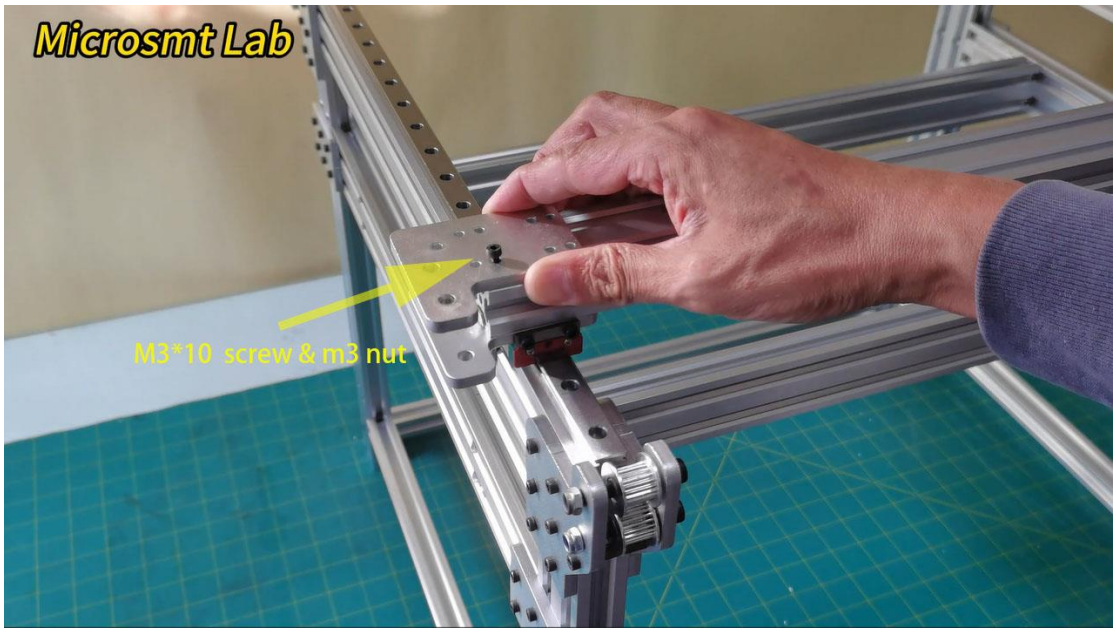

Microsmt Lab

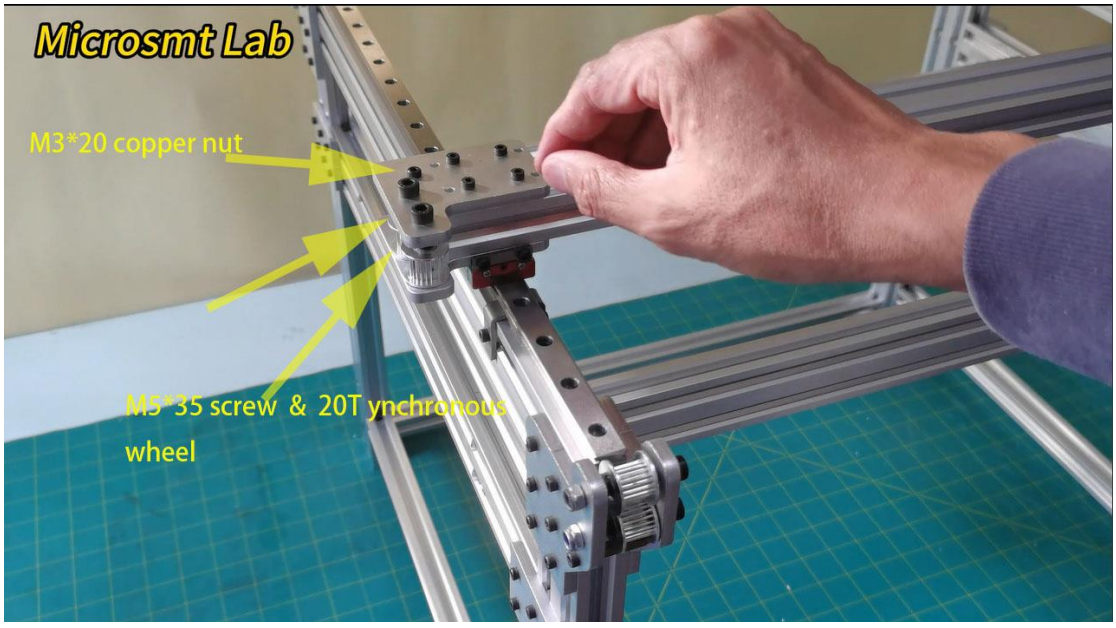

Microsmt Lab

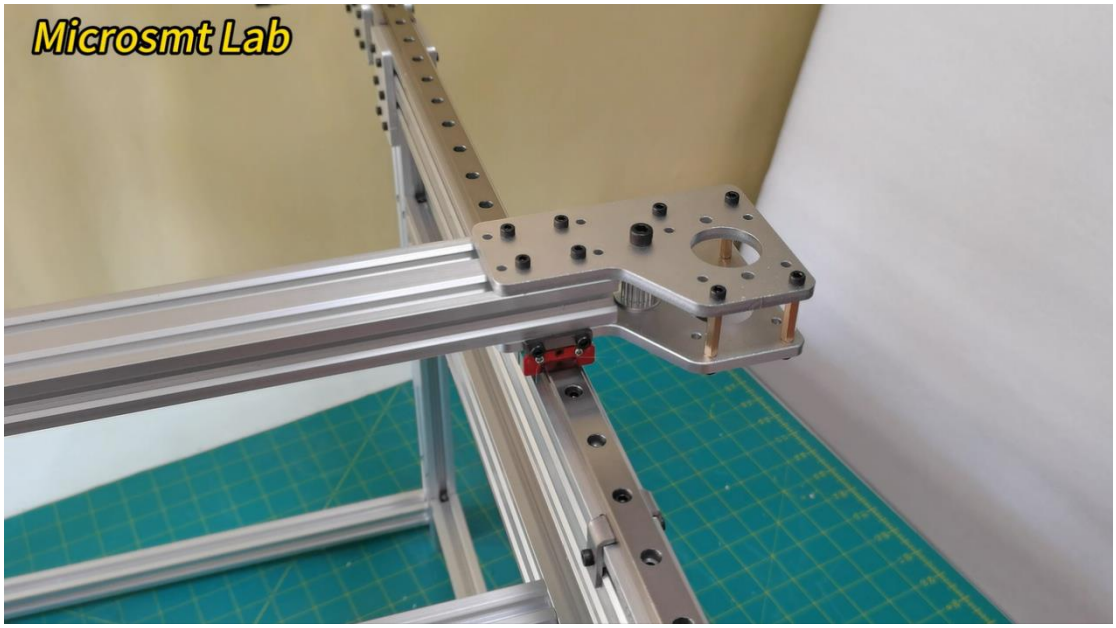
Step 24: Install the 20T synchronous wheel and M3 * 20 copper nut for the X-axis.



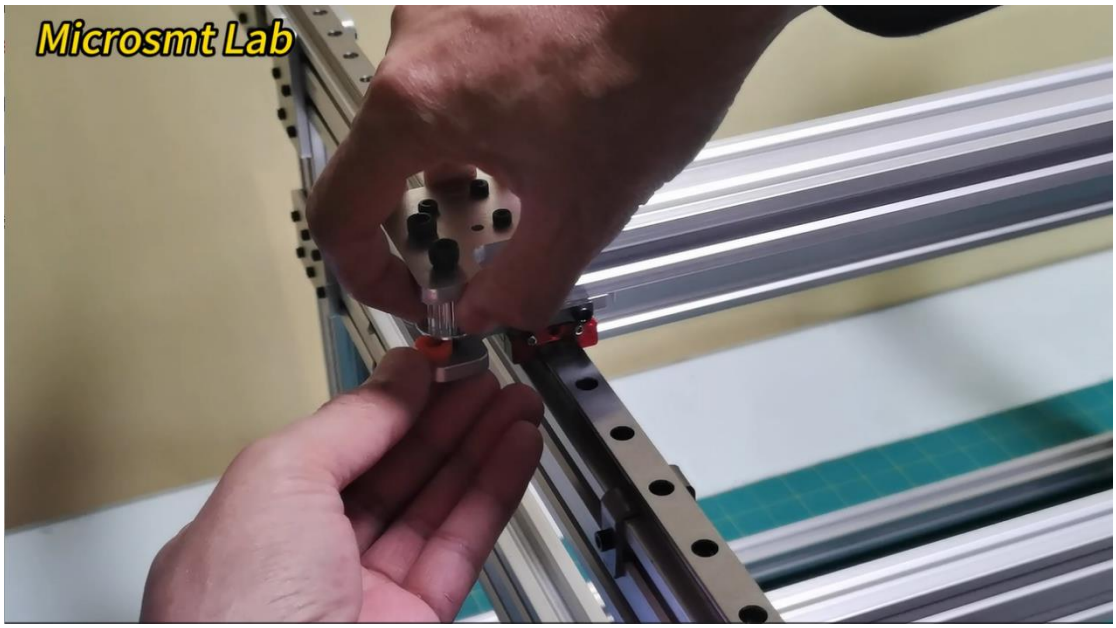


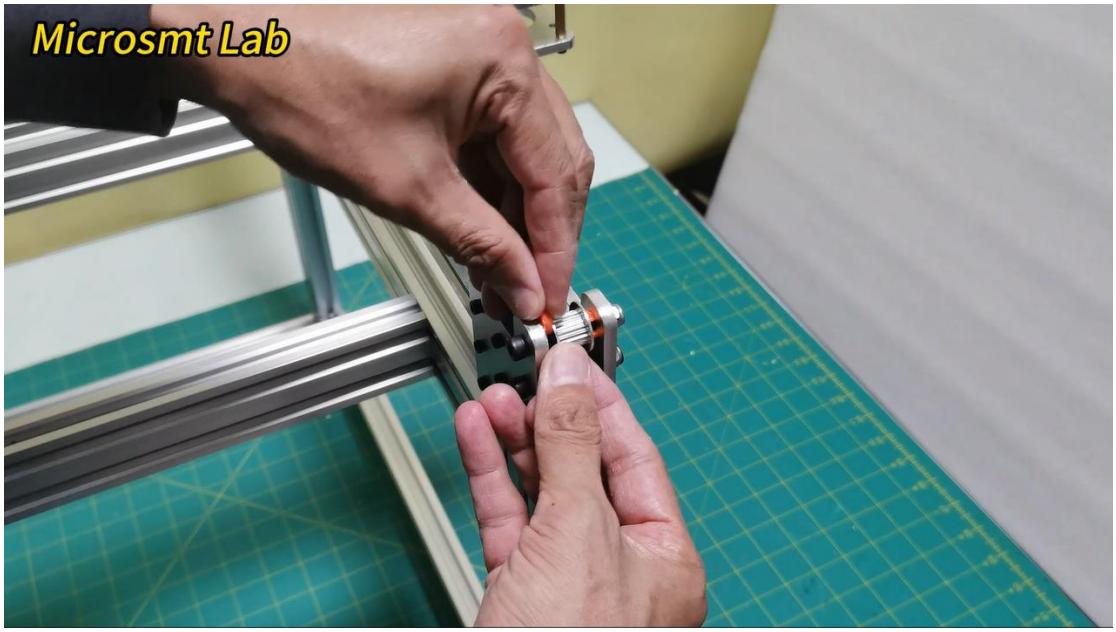

Microsmt Lab
M3*10 screw & m3 nut


Microsmt Lab
M3*20 copper nut
M5*35 screw & 20T ynchronous
wheel

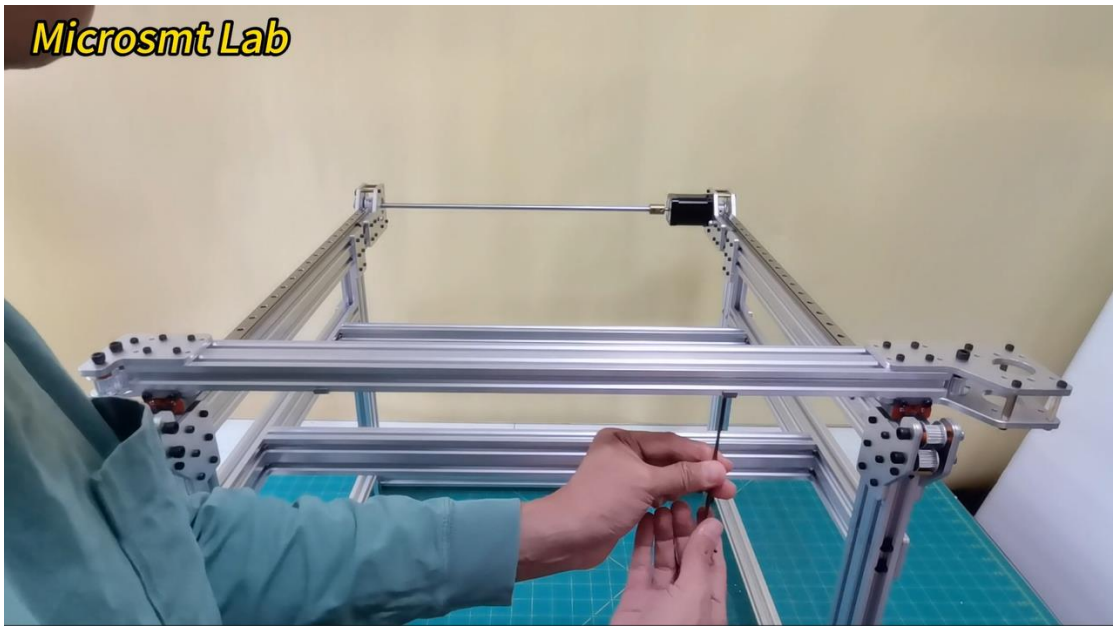


Step 25: Install synchronous wheel limit washer.

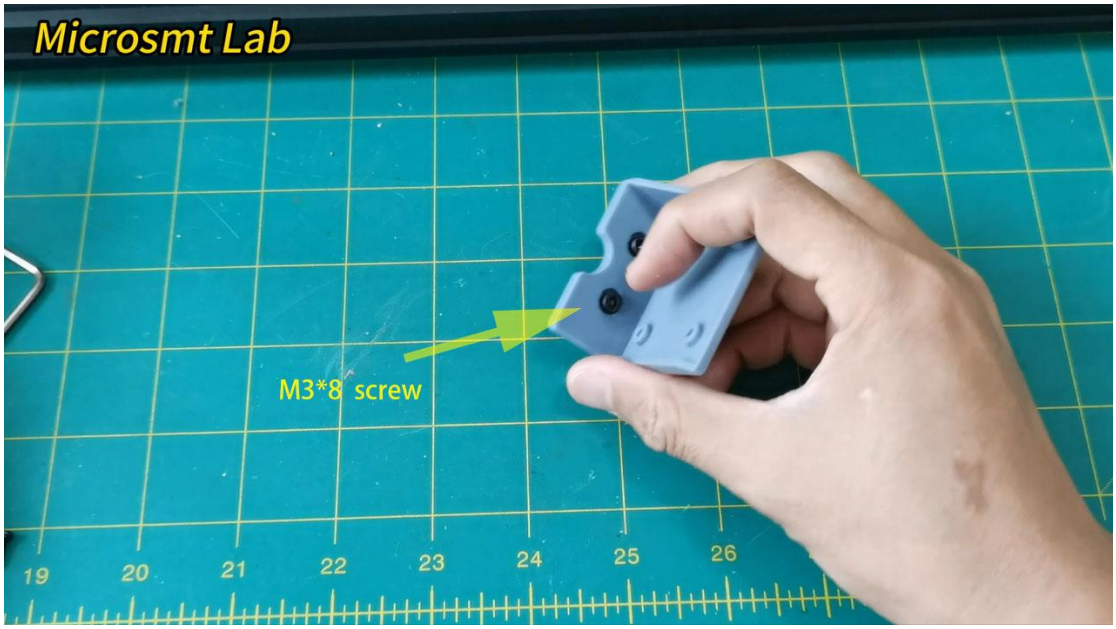




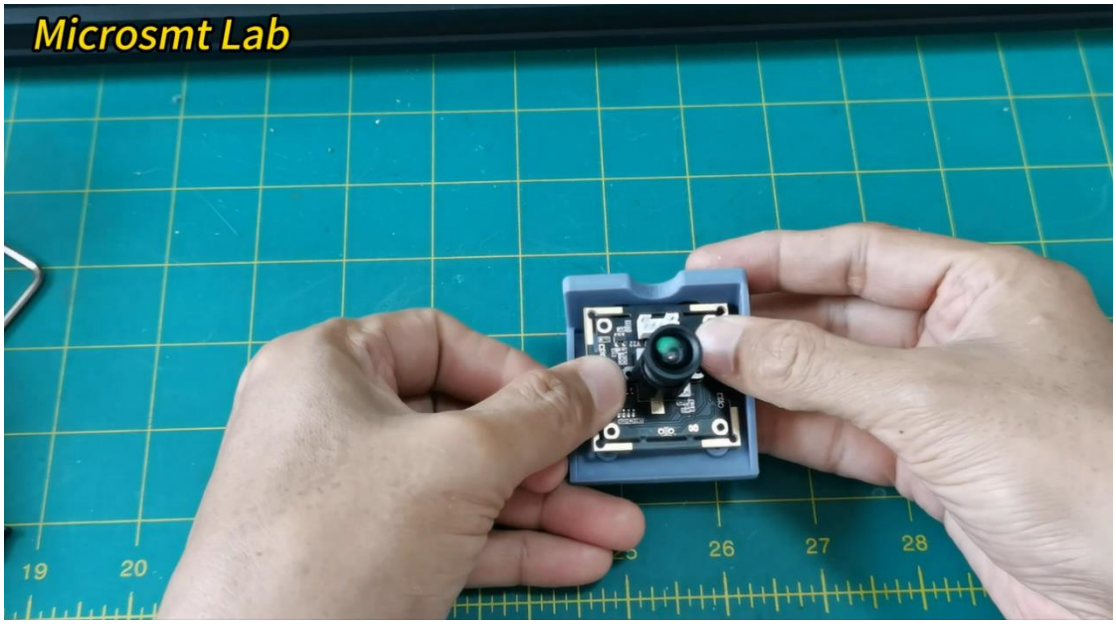
Step 26: Install the limit steel plate for the x-axis linear guide rail.

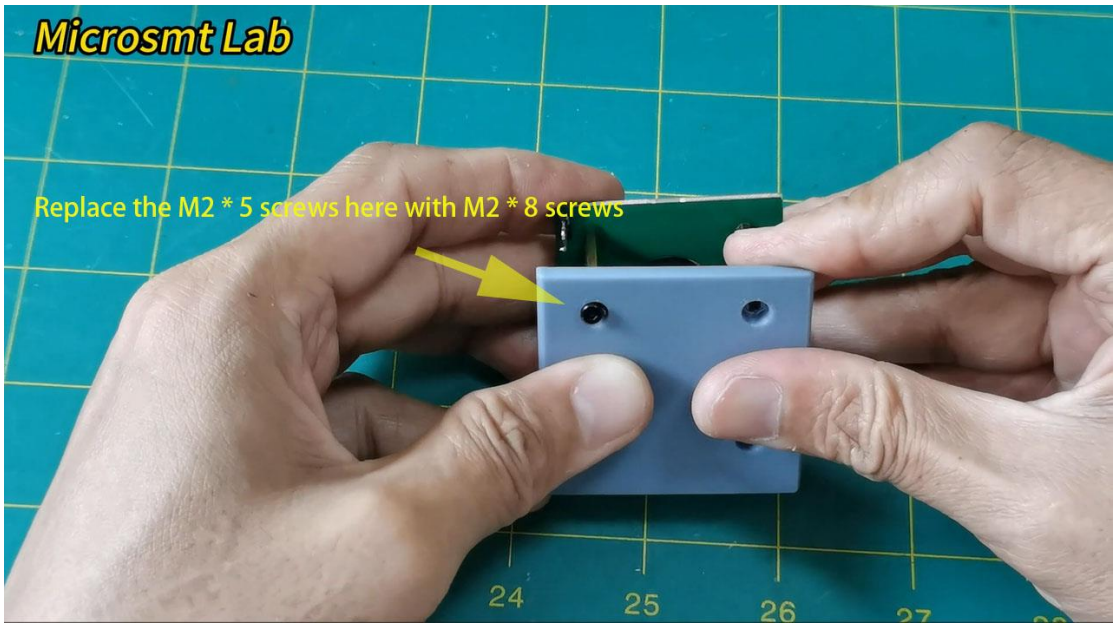


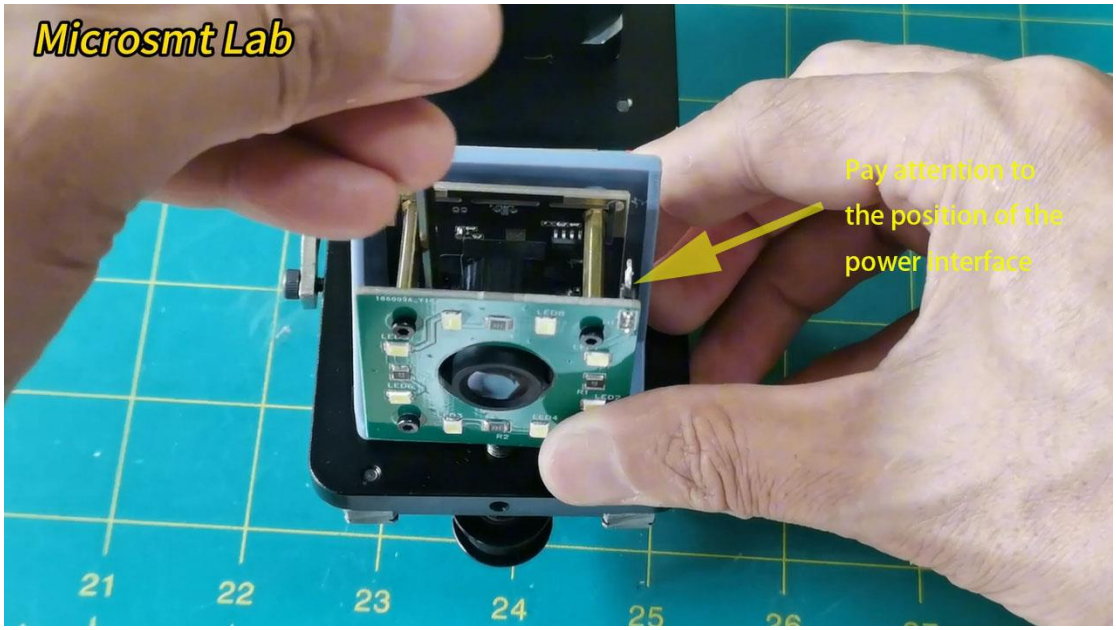
Step 27: Install the top camera.



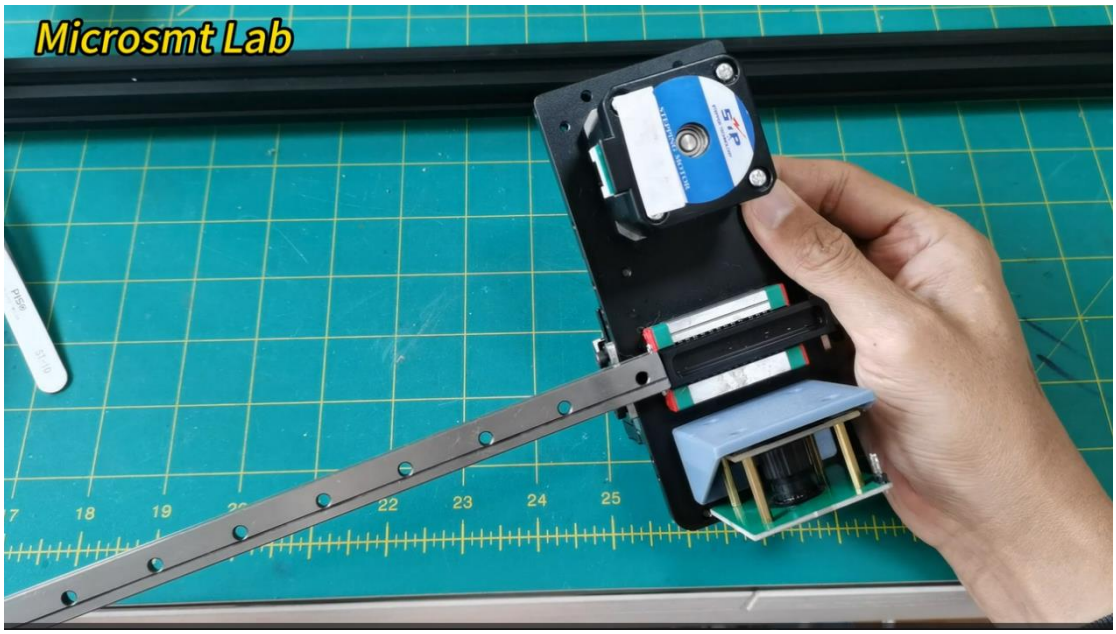
The direction of the camera must be the same as in the photo.

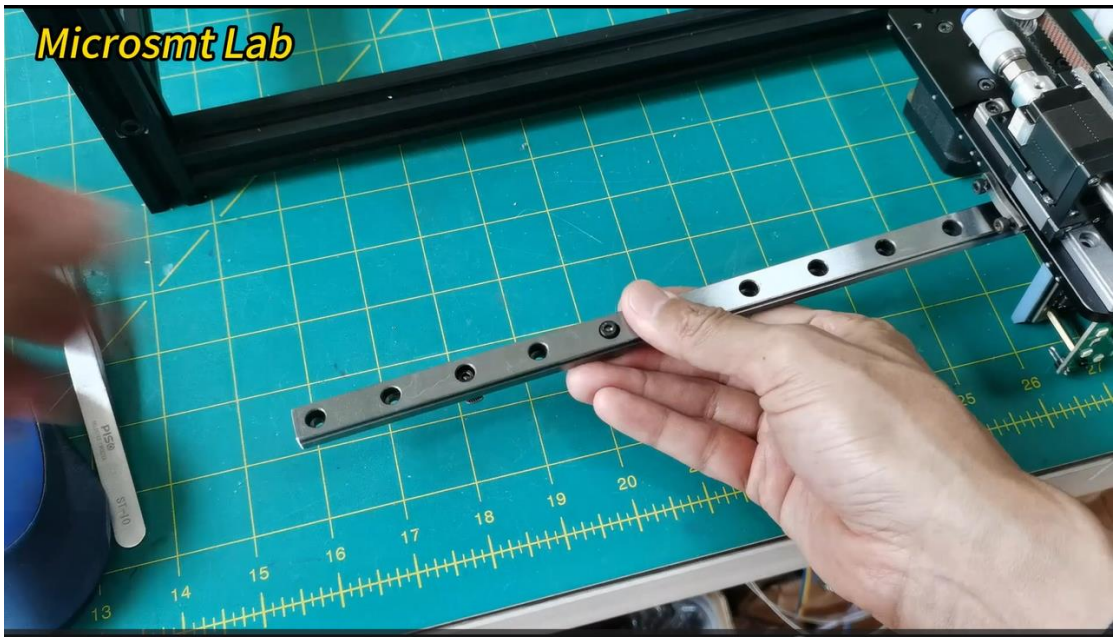


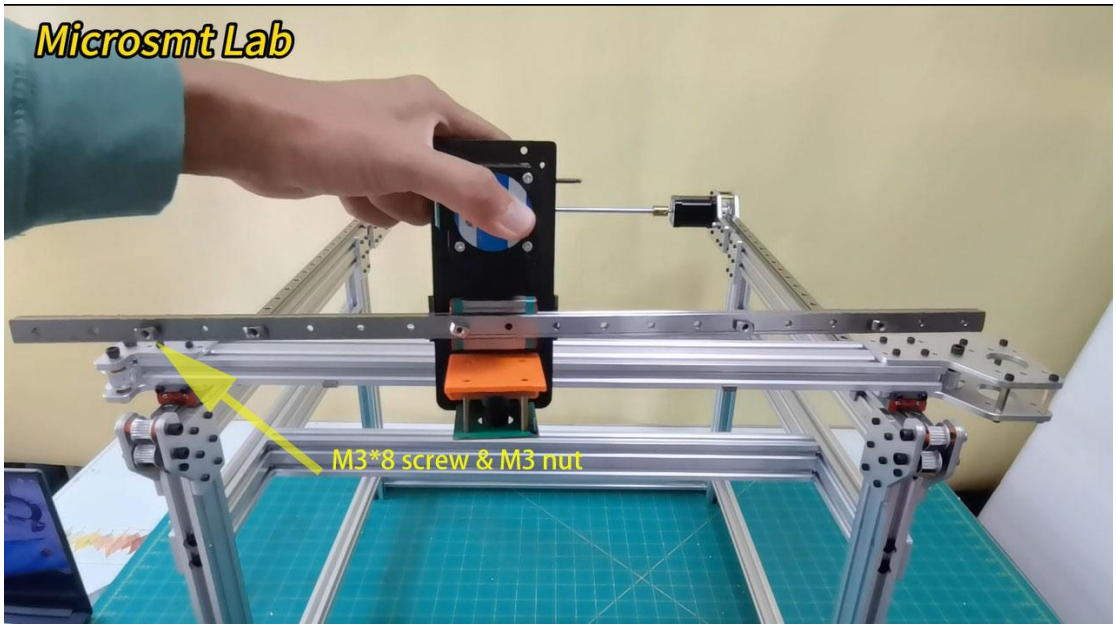

Microsmt Lab
Replace the M2 * 5 screws here with M2 * 8 screws

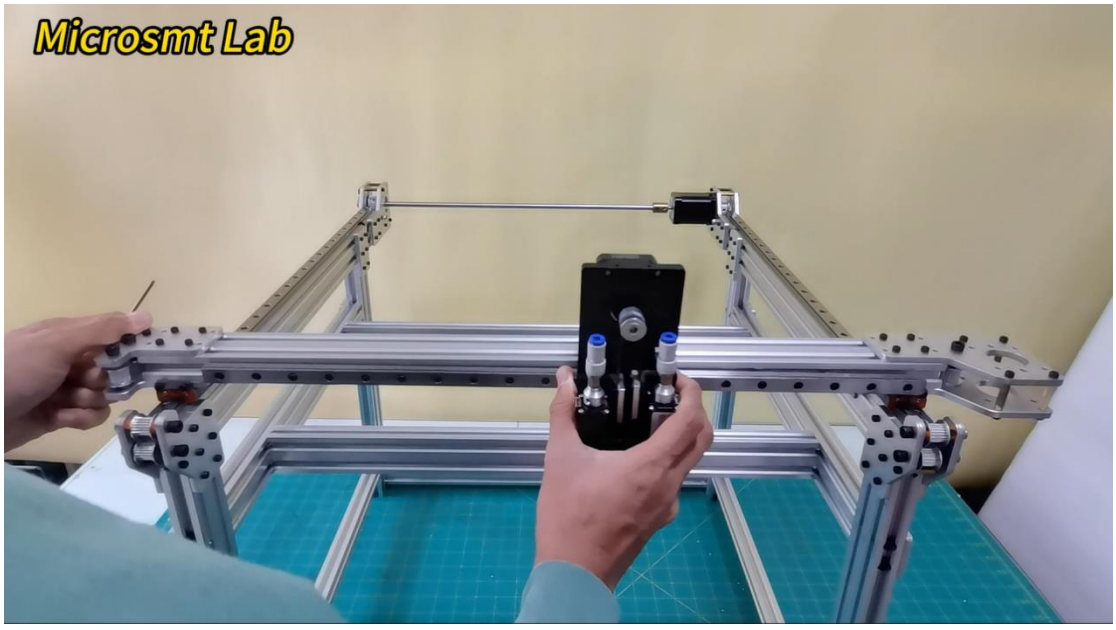

Microsmt Lab
Pay attention to
the position of the
power interface

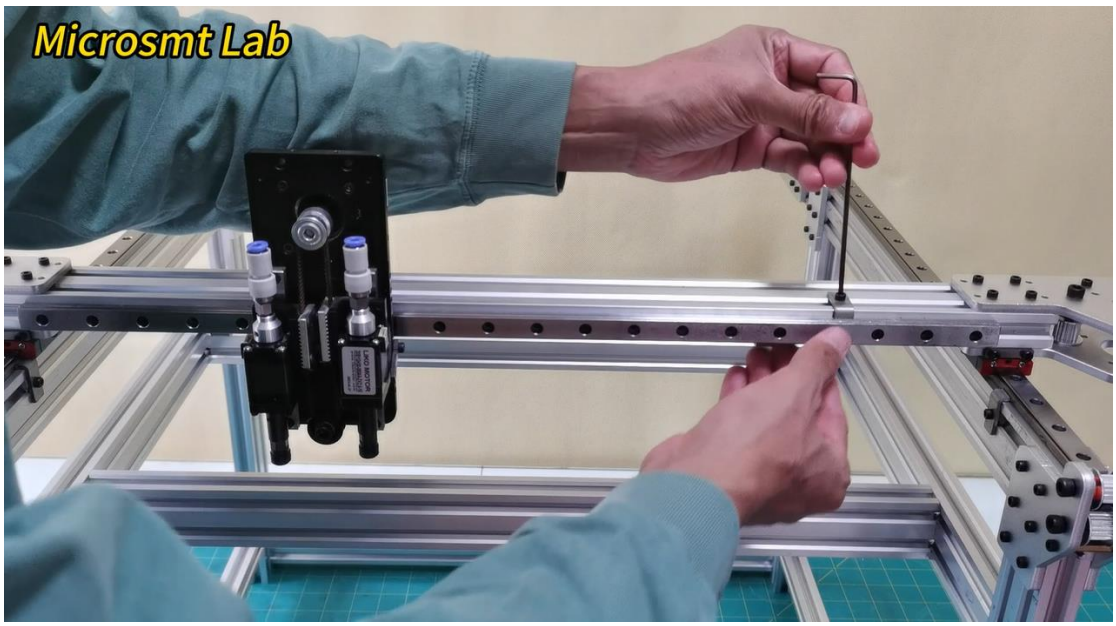
Step 28: Install X-axis linear guide rail.



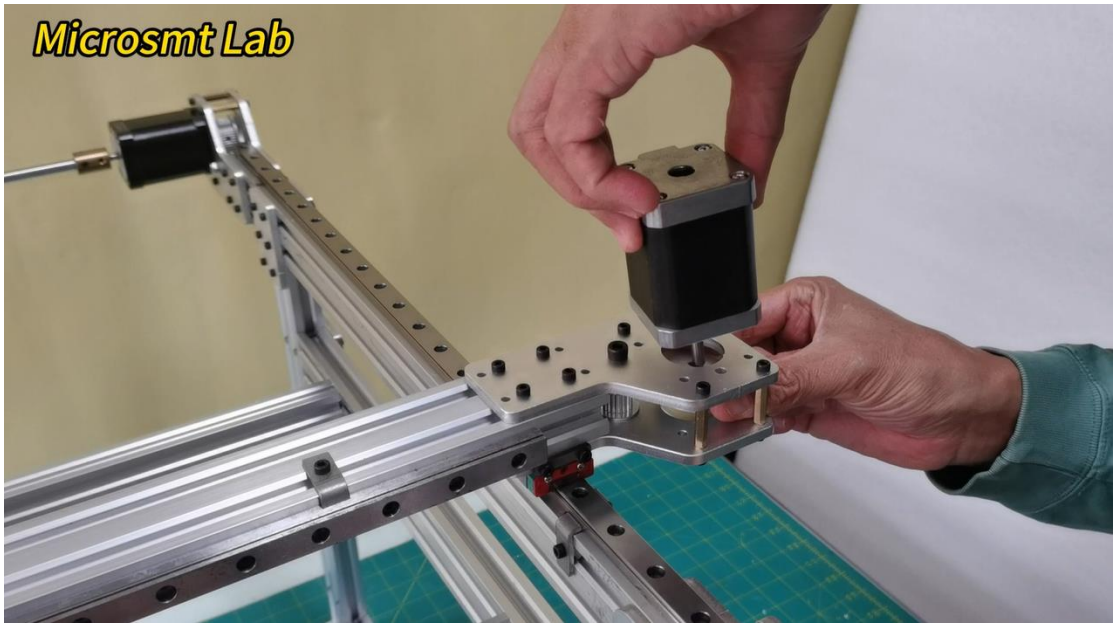



Microsmt Lab
M3*8 screw & M3 nut

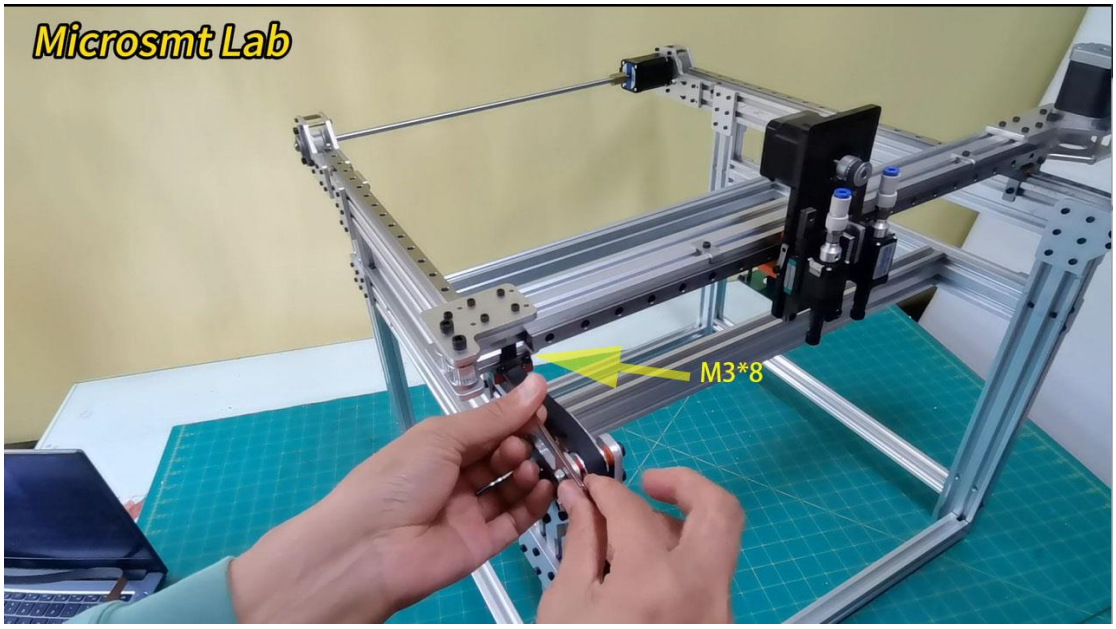

Microsmt Lab

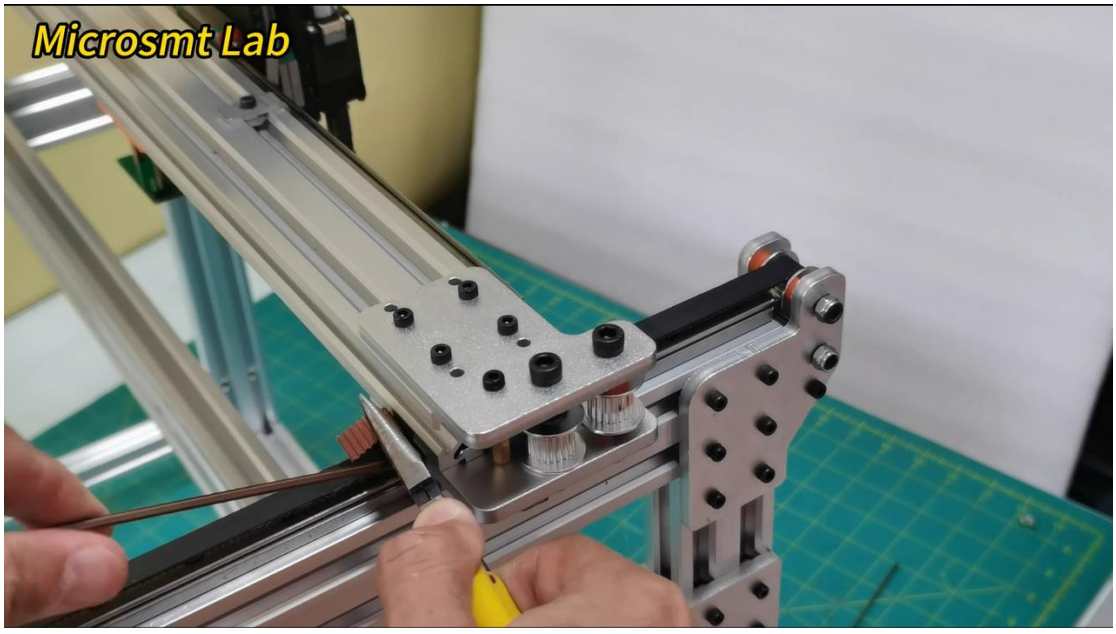


Step 29: Install X-axis motor and 40T synchronous wheel.

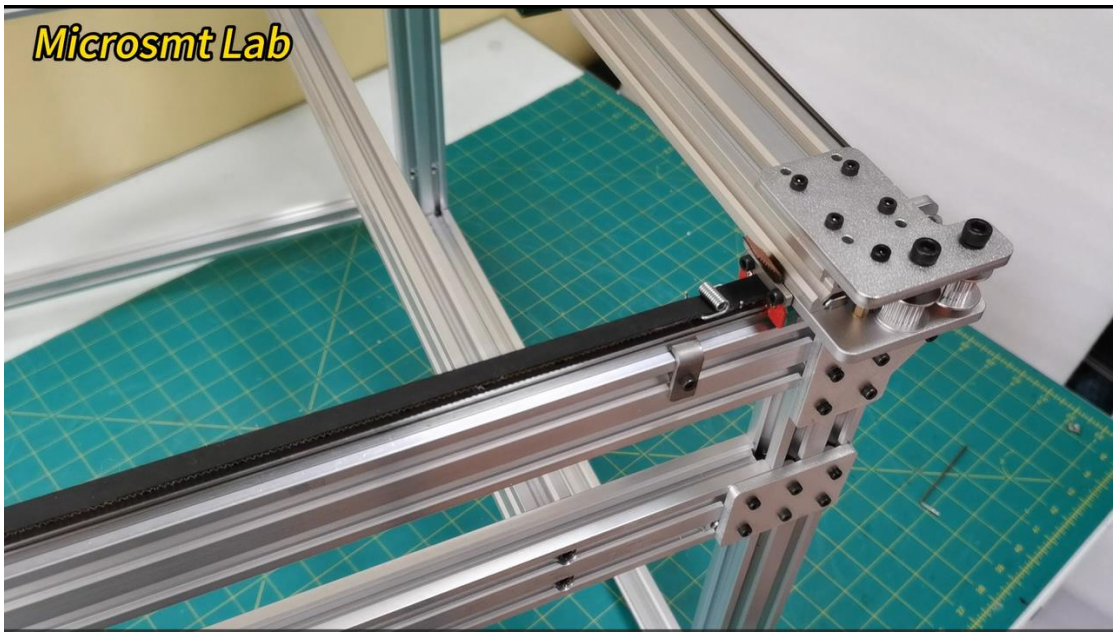


Step 30: Install the y-axis belt.

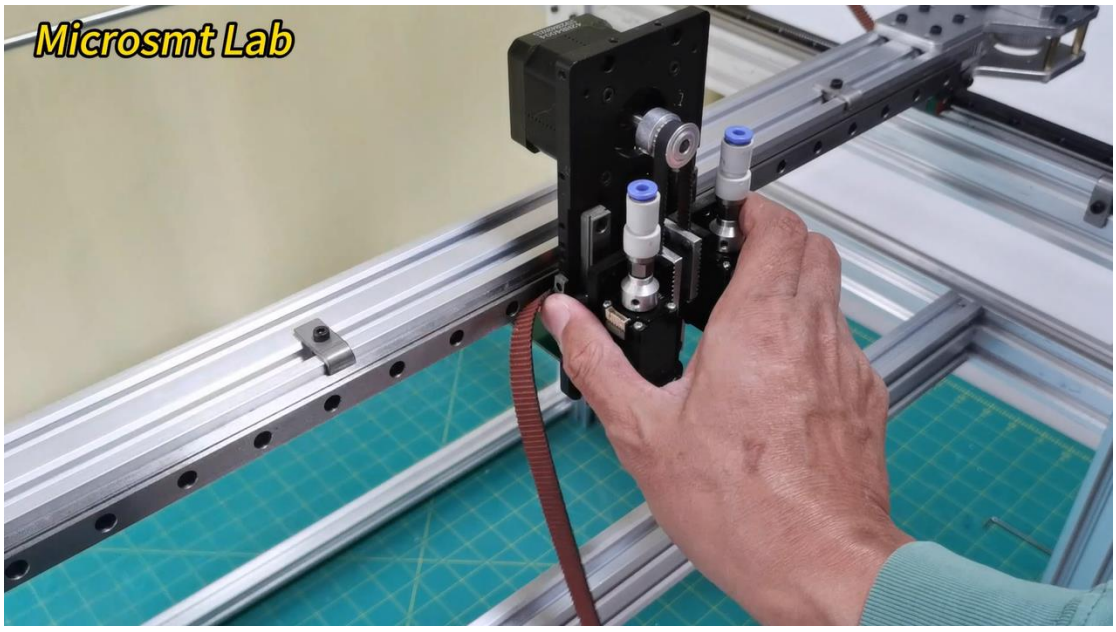


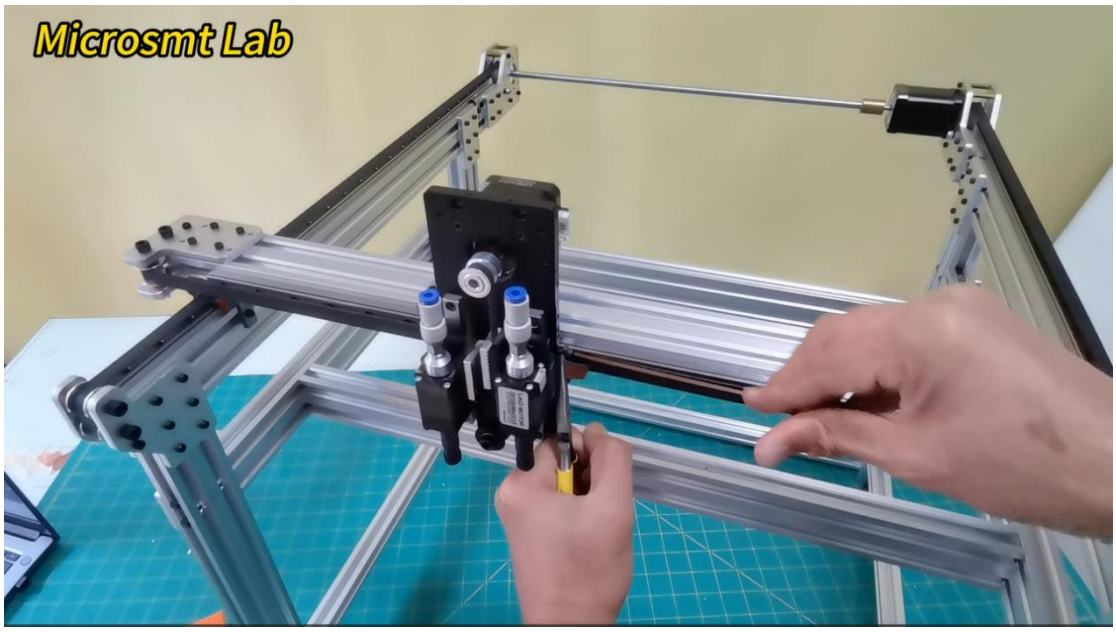


Tighten the belt with a spring.



Step 31：Install the X-axis belt.




Microsmt Lab


Microsmt Lab

Complete the assembly of the frame.

